FIBONACCI–PELL NEAREST GAPS: AN ALL-EXPONENT CLASSIFICATION AND QUADRATIC-UNIT ORBIT RIGIDITY

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Abstract. Let $\lambda = 1 + \sqrt{2}$, let (F_q) be the Fibonacci sequence, and put $D_{q,\sigma} = \sigma F_q + F_{q+1}\sqrt{2}$ for $\sigma \in \{1, -1\}$. For all $q \geqslant 1$ and all positive n, we classify the strict factor-two gaps $\Delta = \lambda^n - D_{q,\sigma}^2$ whose norm is minus the square of their content g. Exactly two occur: $(q, \sigma, n, g, \Delta/g) = (2, -1, 2, 6, \lambda^{-1})$ and $(4, 1, 6, 40, \lambda)$. In particular, no solution has odd exponent. The underlying even windows, by contrast, occur with explicit positive densities governed by an irrational circle rotation.

The argument is unbounded in q; the exact computation enters only after certified analytic estimates have cut it to a finite domain. In the even branch, a primitive factor rectangle separates heights, giving $h < n < 2q$ for the odd exponent $h \geqslant 3$ of the normalised gap $\Delta = g\lambda^{-h}$. Two logarithmic forms give $h = O(\log q)$ and $q < 10^{32}$, certified rational-interval reductions give $h < 192$ and $q < 90$, and exact arithmetic finishes. The odd branch has no such separation, so a fixed-form-first bootstrap is required; it reduces to $q < 96$, where the terminal list is empty.

The map $\mathsf{S}(u, v) = (v, u + 2v)$ is integrally conjugate to multiplication by λ through $A_{(u,v)} = (v - u) + u\sqrt{2}$. This gives an exact orbit/gap equivalence and, for the two Fibonacci cores $(F_{q+1}, F_{q+1} + \sigma F_q)$, a complete all-anchor classification: one persistent family and three isolated time-zero hits are the only Pell-orbit intersections. An exact local rank criterion makes compatible split-prime clocks recur to arbitrary depth, sharply obstructed at 241: congruences alone cannot replace the norm–content equation. The integral Cohn monodromy along every fixed-entry Markoff branch has a quadratic Pisot-unit eigenvalue. On the two constant-directive arms, the golden ratio and λ give explicit integral half-steps on the two-square data. Any analogous half-step on a branch fixing m_0 requires $3m_0 - 2$ to be a square; a finite scan finds one further value, $m_0 = 34$, whose branch unit has an algebraic-unit square root. The two-square sequence on one displayed fixed-34 ray has no common integral half-step. No global two-branch classification is claimed. An occurrence-induced two-square writing also recovers its Farey index and Euclidean cost exactly. Weak minimality of these writings would imply Markoff–Frobenius uniqueness, but that premise remains open; no reduction of a hypothetical Markoff collision to the Fibonacci-core norm–content system is claimed.

1. Introduction

How often can the square of a Fibonacci quadratic integer lie just below a power of the fundamental Pell unit, with the small difference pointing in a primitive negative-norm unit direction? This question combines an Archimedean nearest-neighbour condition with an exact norm–content condition. The former occurs with positive density; the latter will leave only two rows.

Let

$$\lambda = 1 + \sqrt{2}, \qquad D_{q,\sigma} = \sigma F_q + F_{q+1}\sqrt{2}, \qquad q \geqslant 1, \qquad \sigma \in \{1, -1\}. \tag{1}$$

Suppose that a positive integer n lies in the strict factor-two window

2020 *Mathematics Subject Classification.* Primary 11B39, 11D61; Secondary 11J86, 11D09.

Key words and phrases. Fibonacci numbers, Pell equations, quadratic units, linear forms in logarithms, p-adic valuations, arithmetic dynamics, Markoff uniqueness conjecture.

$$D_{q,\sigma}^2 < \lambda^n < 2D_{q,\sigma}^2. \tag{2}$$

There is at most one such n, because the ratio of two distinct integral powers of λ is at least $\lambda > 2$. Write

$$\Delta_{q,\sigma} = \lambda^n - D_{q,\sigma}^2, \qquad g_{q,\sigma} = \operatorname{cont}(\Delta_{q,\sigma}), \tag{3}$$

where

$$N(a + b\sqrt{2}) = a^2 - 2b^2$$

is the quadratic norm and the coefficient content of $a + b\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ is $\gcd(|a|, |b|)$. The target condition is

$$N(\Delta_{q,\sigma}) = -g_{q,\sigma}^2. \tag{4}$$

After division by its largest rational coefficient factor, the gap is then a unit of norm -1. The two surviving examples already display the two possible orientations:

$$\lambda^2 - (-F_2 + F_3\sqrt{2})^2 = -6 + 6\sqrt{2} = 6\lambda^{-1}, \tag{5}$$

$$\lambda^6 - (F_4 + F_5\sqrt{2})^2 = 40 + 40\sqrt{2} = 40\lambda. \tag{6}$$

Our first main theorem says that there are no others, even when the parity of n is not prescribed.

Theorem 1.1 (All-exponent nearest-gap classification)**.** *For $q \geqslant 1$, $\sigma \in \{1, -1\}$, and positive n, the two conditions (2) and (4) hold exactly in the two rows*

q	σ	n	$g_{q,\sigma}$	$\Delta_{q,\sigma}/g_{q,\sigma}$
2	-1	2	6	λ^{-1}
4	1	6	40	λ

(7)

In particular, no odd exponent is a target. Every hypothetical target obeys the intrinsic selector

$$n \textit{ odd} \Longrightarrow 3 \mid q, \qquad n \textit{ even} \Longrightarrow 3 \nmid q. \tag{8}$$

The boundary $q \geqslant 1$ is sharp: at $q = 0$ one has $D_{0,\sigma} = \sqrt{2}$ and $\lambda - D_{0,\sigma}^2 = \lambda^{-1}$.

The proof is an exact, finite certificate built on an unbounded argument. The parity selector first divides the proof into two genuinely different branches. For even n, algebraic reconstruction turns the coefficient content into a primitive factor rectangle. This gives the missing height separation $h < n < 2q$ for the negative unit exponent $-h$. Two linear forms in logarithms then give $h = O(\log q)$ and $q < 10^{32}$; certified rational-interval reductions yield $h < 192$ and $q < 90$, after which exact recurrence arithmetic is exhaustive. For odd n, the selector forces $3 \mid q$. Its factorisation has no analogue of $h < n$, so the proof must be reordered: a fixed-coefficient logarithmic form first bounds $\min\{h, q \log_\lambda \varphi\}$, where $\varphi = (1 + \sqrt{5})/2$, and only then can the moving-coefficient form bound q. The same certified reductions give $h < 48$ or $q < 96$, and the exact terminal list is empty.

The scarcity in theorem 1.1 does not come from scarcity of windows. Theorem 2.3 proves that both even candidate windows occur for natural density $\log_\lambda \sqrt{2} - \eta = 0.3300\ldots$, with η the phase constant of equation (29),, and that the density statement persists on every arithmetic progression. The exact norm–content condition, not the Archimedean approximation, is the rigid step.

Our second main theorem packages the classification as a complete arithmetic orbit result. Let

$$\mathsf{S}(u,v) = (v, u+2v), \qquad \mathcal{H}(u,v) = u^2+v^2, \qquad z_{q,\sigma} = (F_{q+1}, F_{q+1}+\sigma F_q), \tag{9}$$

and write $\lambda^r = C_r + P_r\sqrt{2}$. We call d the *anchor*: it is the Pell index at time $t=0$ in the orbit equation below.

Theorem 1.2 (Complete Fibonacci-core Pell-orbit classification)**.** *For every $q \geqslant 1$, $\sigma \in \{1,-1\}$, positive anchor d, and $t \geqslant 0$, one has*

$$\mathcal{H}(\mathsf{S}^t z_{q,\sigma}) = P_{d+2t} \tag{10}$$

if and only if one of the following alternatives holds:

σ	q	d	t
1	1	3	*arbitrary* $t \geqslant 0$
-1	1	1	0
-1	2	3	0
-1	4	5	0

(11)

There is no hit at a positive even anchor. In particular, the consecutive-Fibonacci core $z_{q,1} = (F_{q+1}, F_{q+2})$ has no hit for $q \geqslant 2$.

The bridge behind this theorem is more general than the Fibonacci cores. For a positive ordered integral seed $z=(u,v)$, the map

$$A_z = (v-u) + u\sqrt{2} \tag{12}$$

obeys $A_{\mathsf{S}^t z} = \lambda^t A_z$. Thus an integral linear dynamical system is integrally conjugate to multiplication by a quadratic unit. An orbit hit at anchor d is equivalent, outside the canonical unit seeds, to the anchored gap

$$\lambda^{d-1} - A_z^2 = g\lambda^{-(2t+1)}, \tag{13}$$

which automatically lies in the strict factor-two window. This equivalence retains the anchor and time in both directions; it is not merely a numerical analogy.

Several classical structures clarify why the proof works. The equations $N(X+Y\sqrt{2}) = \pm 1$ define the norm-one torus and the torsor of norm -1, so coefficient content records the integral lattice multiple of a primitive torsor point. The matrix of S lies in $\mathrm{GL}_2(\mathbb{Z})$, and its symmetric-square action preserves the Lorentzian identity $2\mathcal{H}^2 - \mathcal{B}^2 - \mathcal{Q}^2 = 0$ used in the orbit bridge. Two of the three involutions of the biquadratic field $\mathbb{Q}(\sqrt{2},\sqrt{5})$ carry the argument: one controls Pell norms, while the other separates the dominant and conjugate Fibonacci terms in the logarithmic forms. Finally, the local results combine p-adic rank lifting with an Archimedean irrational rotation. These are operational connections: each supplies an invariant, factorisation, height, or recurrence mechanism used below.

Interactions between the Fibonacci and Pell sequences have been studied from several directions. Alekseyev determined the exact intersection of the two sequences [2], and Bravo, Herrera and Luca extended the question to common values of their generalised analogues [5]. Pomeo and Bravo measured how closely a Pell number can approach a Fibonacci number [17], Luca and Zottor classified the Fibonacci numbers occurring among the Y-coordinates of Pell equations [15], and Kafle, Luca, Montejano, Szalay and Togbé treated X-coordinates that are products of two Fibonacci numbers [11]. Each of those questions asks for a value coincidence between two sequences, or for an Archimedean bound on their difference. The present question is different in kind. The window (2) is only its Archimedean part, and the arithmetic lies in the exact norm–content equation (4) imposed on the gap itself. That equation sees both coefficient coordinates of $\Delta_{q,\sigma}$ at once, which is why a positive-density supply of windows leaves exactly two rows.

The nearest-gap problem arose in an ongoing programme on the Markoff–Frobenius uniqueness conjecture. A positive integral triple (a, b, c) with $a \leqslant b \leqslant c$ is a Markoff triple when

$$a^2 + b^2 + c^2 = 3abc, \tag{14}$$

and the conjecture asks whether the largest entry c determines the unordered pair $\{a, b\}$. The problem has arithmetic, geometric, word-combinatorial, and trace formulations; see Aigner [1], the integral-necklace model of Fisac [8], and Reutenauer's account of Christoffel words and Markoff numbers [19]. The present paper neither proves nor disproves this conjecture. In particular, we do not prove that a hypothetical Markoff collision must enter the Fibonacci-core family in theorem 1.2. The arithmetic theorem is complete on its stated domain; the upstream reduction remains open.

For any fixed factored value M, all primitive two-square writings and their candidate Euclidean and Farey itineraries can be enumerated exactly. As remark 10.2 explains, this finite localization does not supply the uniform obstruction needed to rule out two distinct valid occurrences for every M.

The motivation does, however, distinguish the two sequences, and section 9 explains why. Along each branch that keeps one Markoff entry m_0 fixed, the integral unimodular Cohn monodromy has dominant eigenvalue equal to the quadratic Pisot unit that solves $X^2 - 3m_0X + 1 = 0$. This eigenvalue is not, in general, the value of the corresponding infinite continued fraction. Its conjugate contraction controls the scale on every such branch; it does not single out Fibonacci and Pell. What the orbit bridge of section 7 consumes is stronger: an integral square root acting compatibly on the two-square data. Comparing traces shows that any such half-step forces $3m_0 - 2$ to be a perfect square. The Fibonacci and Pell arms, carrying φ and λ, provide the two explicit actions used here and are the two constant-directive branches. The necessary square condition is not proved to select them globally, and no such global classification is used in the paper.

The local theory also draws a sharp methodological boundary. A natural plus-sign cross-gcd on genuine nearest windows contains arbitrary prescribed powers of 7 and 17, and more generally simultaneous products from an infinite class of split primes. A prime-rank parity criterion is exact, while the split prime 241 gives a sharp obstruction. These facts rule out any proof based only on uniform boundedness, squarefreeness, or finitely many compatible local rank clocks. They do not produce a norm–content target. Figure 1 displays the dependency structure and the one unproved Markoff-facing arrow.

The paper is organised as follows. Section 2 develops the quadratic-unit dictionary, the exact window densities, and the primitive factor rectangle. Sections 3 to 5 prove the even branch, and section 6 proves the odd exclusion. Section 7 gives the general seed/unit integral conjugacy and the complete Fibonacci-core orbit theorem. Section 8 develops the local rank criterion and simultaneous prime-power recurrence. Section 9 explains the Pisot monodromy scales and half-step obstruction. Section 10 reconstructs the exact Farey cost from a valid two-square writing and isolates weak two-square minimality as a sufficient but open route to Markoff–Frobenius uniqueness. We close by stating the exact Markoff-facing boundary and further directions. Section A specifies the finite certificates and their coverage.

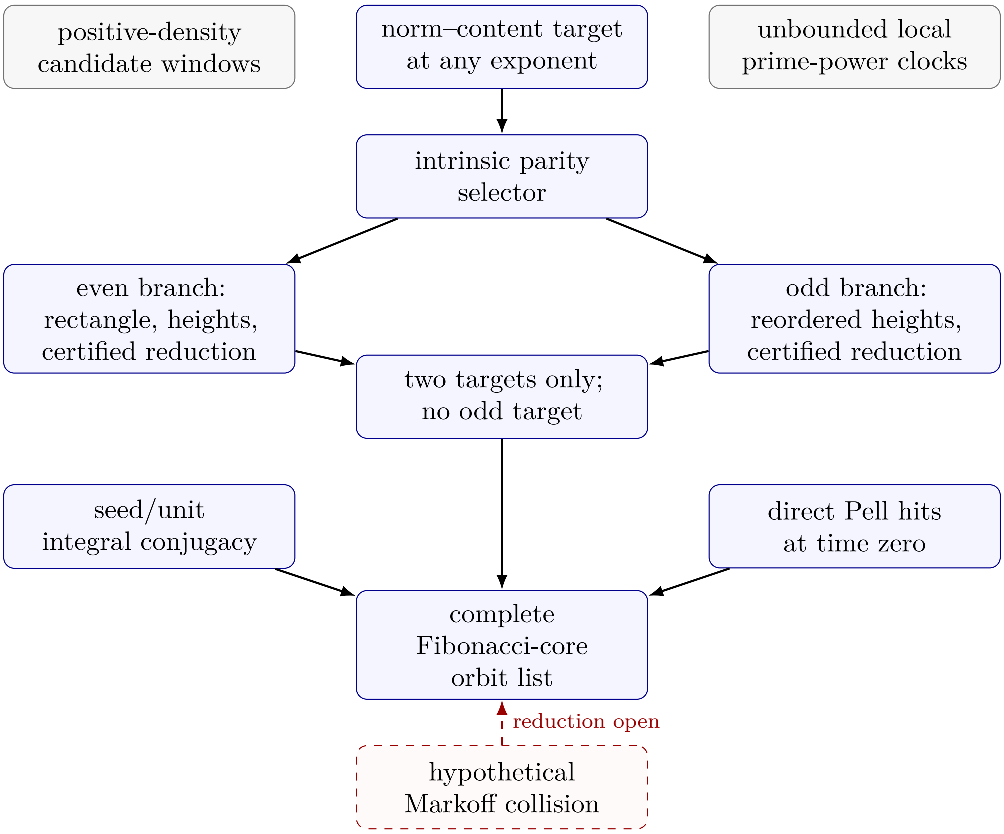

FIGURE 1. Logical architecture. Solid arrows are proof dependencies: the orbit list combines the independent seed/unit conjugacy, the all-exponent classification, and the direct time-zero Pell hits. Density and local clocks are contextual, not hypotheses. The dashed red arrow is the unproved Markoff-to-orbit reduction.

2. Quadratic units and the intrinsic nearest-gap system

This section builds the arithmetic dictionary used throughout the paper. A candidate window is an Archimedean inequality; a target is a candidate whose two coefficient coordinates also satisfy the exact norm–content equation. Keeping those two levels separate is essential.

2.1. **Quadratic integers and primitive-unit normalisation.** We use the Fibonacci sequence

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{r+2} = F_{r+1} + F_r, \tag{15}$$

and the Pell coordinates defined for every $r \in \mathbb{Z}$ by

$$\lambda^r = C_r + P_r\sqrt{2}, \qquad \lambda = 1 + \sqrt{2}. \tag{16}$$

Thus $P_0 = 0$, $P_1 = 1$, $P_{r+2} = 2P_{r+1} + P_r$, and $C_r = P_r + P_{r-1}$ for $r \geqslant 1$. Since $\overline{\lambda} = 1 - \sqrt{2} = -\lambda^{-1}$,

$$C_r^2 - 2P_r^2 = (-1)^r. \tag{17}$$

Multiplication by λ^r acts on coefficient columns by the integral matrix

$$\begin{pmatrix} C_r & 2P_r \\ P_r & C_r \end{pmatrix}, \tag{18}$$

whose determinant is $(-1)^r$. It therefore preserves coefficient content. Equivalently, for $\alpha \in \mathbb{Z}[\sqrt{2}]$, $\mathrm{cont}(\alpha)$ is the largest positive integer c such that $\alpha \in c\mathbb{Z}[\sqrt{2}]$, and multiplication by a unit is a unimodular change of the coefficient lattice.

There is also a useful arithmetic-geometric reading. The Pell conic $X^2 - 2Y^2 = 1$ is the norm-one torus for $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$, while $X^2 - 2Y^2 = -1$ is its norm--1 torsor. The coefficient-primitive integral points on the latter with positive principal embedding are precisely λ^j for odd $j \in \mathbb{Z}$. Thus the target condition below says that the primitive direction of a gap lands on this torsor; see, for example, the Pell-conic viewpoint in [13, Chapter 7].

More explicitly, the regular representation

$$\iota(a + b\sqrt{2}) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}, \qquad \det \iota(a + b\sqrt{2}) = a^2 - 2b^2, \tag{19}$$

identifies the integral points of the norm-one torus with the rank-one arithmetic group

$$T(\mathbb{Z}) = \{\pm\lambda^{2m} : m \in \mathbb{Z}\}. \tag{20}$$

The norm--1 integral fibre is the coset $\lambda T(\mathbb{Z})$. Hence a target gap is a rational lattice multiple of a point in this coset, and division by the coefficient content recovers its primitive arithmetic-group direction. This also explains why the matrix in equation (18) preserves content: it is the integral regular representation of a unit.

For the rest of the paper let

$$f = F_q, \qquad x = F_{q+1}, \qquad k = R_q = 2x^2 - f^2, \tag{21}$$

and define

$$A_q = f + x\sqrt{2}, \qquad B_q = x\sqrt{2} - f. \tag{22}$$

For $q \geqslant 1$, both numbers are positive and $A_q B_q = k$. The two signs in equation (1) are simply $D_{q,1} = A_q$ and $D_{q,-1} = B_q$.

Lemma 2.1 (Uniqueness and unit normalisation)**.** *For a fixed pair (q, σ) with $q \geqslant 1$, at most one positive integer n satisfies*

$$D_{q,\sigma}^2 < \lambda^n < 2D_{q,\sigma}^2. \tag{23}$$

If this integer exists, put $\Delta = \lambda^n - D_{q,\sigma}^2$ and $g = \mathrm{cont}(\Delta)$. Then

$$N(\Delta) = -g^2 \tag{24}$$

if and only if

$$\Delta = g\lambda^j \tag{25}$$

for a unique odd integer j. In that case $M = n - j$ is positive and $M \equiv n - 1 \pmod 2$.

Proof. The ratio of two distinct integral powers of λ is at least $\lambda > 2$, proving uniqueness. If equation (24) holds, then Δ/g is a coefficient-primitive algebraic integer of norm -1 and positive principal embedding. The unit group of $\mathbb{Z}[\sqrt{2}]$ gives $\Delta/g = \lambda^j$ for one odd j. The converse is immediate from equation (17). Finally, $\lambda^j = \Delta/g < \lambda^n$, so $j < n$ and $M = n - j > 0$; the congruence for M follows because j is odd. ∎

Proposition 2.2 (Global parity selector)**.** *Every target satisfies*

$$n \text{ odd} \Longrightarrow 3 \mid q, \qquad n \text{ even} \Longrightarrow 3 \nmid q. \tag{26}$$

Proof. The coefficient pair of the gap is

$$(C_n - (f^2 + 2x^2),\ P_n - 2\sigma f x). \tag{27}$$

Modulo 2, one has $C_n \equiv 1$, $P_n \equiv n$, and $f^2 + 2x^2 \equiv f$. If the target condition holds, the quotient by the content is an odd power of λ, whose two coefficients are odd. Consequently the two entries in equation (27) have the same parity, and

$$n \equiv 1 - F_q \pmod 2. \tag{28}$$

The Fibonacci number F_q is even exactly when $3 \mid q$, proving equation (26). □

The selector is a conclusion, not an external admissibility hypothesis. The even branch automatically has $3 \nmid q$, hence f and k are odd and M is odd. Those are precisely the hypotheses used by the primitive reconstruction below. The odd branch has a different factorisation and will be treated in section 6.

2.2. **Candidate windows are abundant.** Before imposing the norm–content equation, it is useful to know how often the Archimedean window itself occurs. This separates the elementary supply of candidates from the arithmetic rigidity that eventually leaves only two targets. Put

$$\vartheta = \log_\lambda \sqrt{2}, \qquad \eta = \log_\lambda \left(\frac{\sqrt{2}\varphi + 1}{\lambda(\sqrt{2}\varphi - 1)} \right), \qquad \varphi = \frac{1 + \sqrt{5}}{2}. \tag{29}$$

Theorem 2.3 (Exact density of the even candidate windows)**.** *Call the sign σ available at q if an even positive integer n satisfies equation (23). Then*

$$0 < \eta < \vartheta < \frac{1}{2}, \tag{30}$$

and the four sign patterns have natural densities

available signs	*both*	*plus only*	*minus only*	*neither*
density	$\vartheta - \eta$	η	η	$1 - \vartheta - \eta$

(31)

The same relative densities hold after restricting q to any fixed arithmetic progression. In particular, they remain valid on either class $q \equiv 1, 2 \pmod 3$ and on their union $3 \nmid q$.

Proof. Write $x_\sigma(q) = \log_\lambda D_{q,\sigma}$. If $n = 2m$, then equation (23) is equivalent, away from its strict endpoints, to

$$\{x_\sigma(q)\} \in I, \qquad I = (1 - \vartheta, 1). \tag{32}$$

Binet's formula gives

$$x_\sigma(q) = q\gamma + \beta_\sigma + O(\varphi^{-2q}), \qquad \gamma = \frac{\log \varphi}{\log \lambda}, \qquad \beta_\sigma = \log_\lambda \left(\frac{\sqrt{2}\varphi + \sigma}{\sqrt{5}} \right). \tag{33}$$

The number γ is irrational. Indeed, a rational relation would put equal positive powers of φ and λ in $\mathbb{Q}(\sqrt{5}) \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$, whereas no positive power of φ is rational. Hence $q\gamma + \beta_-$ is equidistributed modulo one by Weyl's theorem [24]. Equivalently, its empirical measures converge to Haar probability measure on $\mathbb{R}/\mathbb{Z}$. Since

$$\beta_+ - \beta_- = 1 + \eta, \tag{34}$$

the minus interval is $I_- = (1-\vartheta, 1)$ and the plus interval, expressed in the same phase coordinate, is

$$I_+ = (1 - \vartheta - \eta, 1 - \eta). \tag{35}$$

The endpoint order in figure 2 gives the four lengths in equation (31); these lengths are exactly the Haar measures of the corresponding phase regions. The inequalities in equation (30) reduce respectively to $1+\sqrt{2} > \sqrt{5}$, $\varphi > 3/2$, and $\lambda > 2$. The exponentially decaying error in equation (33) can change membership only near the four endpoints, a set of density zero. On a progression $q = q_0 + Q\ell$, the increment $Q\gamma$ remains irrational, so the same argument applies. ☐

There is an exact probability interpretation of this deterministic theorem. If Q_N is chosen uniformly from $\{1, \ldots, N\}$, then the four availability events converge to the probabilities in equation (31). The probability space is simply the phase circle with Haar measure; no independence assumption or random mechanism enters the arithmetic.

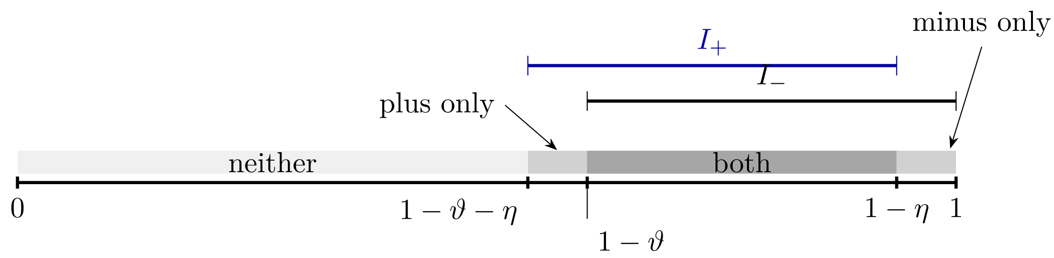

FIGURE 2. The exact phase partition for the two even nearest-window candidates. Irrational rotation distributes the limiting phase uniformly; the four natural densities are therefore the displayed interval lengths. Availability is only an Archimedean condition and does not imply the exact norm–content equation.

Numerically, $\vartheta = 0.3932198506\ldots$ and $\eta = 0.0631959999\ldots$, so both windows occur for about 33% of all indices. The final classification will show that this positive-density supply contains only two norm–content targets. Scarcity of candidate windows is therefore not the exclusion mechanism.

2.3. **Primitive reconstruction in the even branch.** The next reconstruction is the geometric input behind the later height separation. We include the proof because using an arbitrary divisor of $(k^2-1)/4$ here would be invalid.

Lemma 2.4 (Primitive rectangle reconstructed from a target). *Assume that n is even and that equations (23) and (24) hold. With $M = n - j$, there is an odd integer W such that*

$$W^2 = C_M^2 + 1 - k^2, \qquad \gcd(k, W) = 1, \qquad |W| < C_M. \tag{36}$$

Put

$$U = \frac{C_M - W}{2}, \qquad V = \frac{C_M + W}{2}. \tag{37}$$

Then U, V are positive integers of opposite parity and

$$UV = \frac{k^2 - 1}{4}. \tag{38}$$

Moreover, $g/2$ equals U on the plus branch and V on the minus branch, and

$$g < k, \qquad U \geqslant \sqrt{k}, \qquad V \geqslant \sqrt{k}. \tag{39}$$

Proof. On the plus branch, division of $\lambda^n - A_q^2 = g\lambda^j$ by λ^j gives

$$A_q^2 \lambda^{-j} = C_M - g + P_M\sqrt{2}. \tag{40}$$

On the minus branch the same calculation gives

$$B_q^2 \lambda^{-j} = C_M - g + P_M\sqrt{2}. \tag{41}$$

Define $W = C_M - g$ in equation (40) and $W = g - C_M$ in equation (41); thus the rational coefficient in those two equations is, respectively, W and $-W$. The upper half of the strict window gives $D_{q,\sigma}^2 > \lambda^n/2$, and therefore

$$g\lambda^j = \lambda^n - D_{q,\sigma}^2 < \frac{\lambda^n}{2}.$$

Thus $2g < \lambda^{n-j} = \lambda^M$. Since M is odd,

$$\frac{\lambda^M}{2} = C_M + \frac{\lambda^{-M}}{2} < C_M + 1. \tag{42}$$

Thus $g \leqslant C_M$. Equality would give $W = 0$ and $k^2 = C_M^2 + 1$, which is impossible modulo 8 because k and C_M are odd. Therefore equation (36) has $|W| = C_M - g < C_M$.

Taking norms in equations (40) and (41) and using equation (17) proves the first equality in equation (36). To prove primitivity, suppose that an odd prime p divides both k and W. The norm relation then gives $p \mid P_M$. After multiplication by a Pell unit, p would divide both coefficients of A_q^2 or B_q^2. Those coefficients are $f^2 + 2x^2$ and $2fx$ up to signs. Since $\gcd(f, x) = 1$ and $p \mid 2x^2 - f^2$, this is impossible. Hence $\gcd(k, W) = 1$.

Factoring $C_M^2 - W^2 = k^2 - 1$ proves equation (38). Since M and k are odd, the norm relation shows that W is odd. Hence $U + V = C_M$ is odd, so U and V have opposite parities. Put

$$X_\delta = \frac{k - \delta}{2}, \qquad Y_\delta = \frac{k + \delta}{2}. \tag{43}$$

The two factors X_δ and Y_δ are coprime and consecutive. Choose the unique $\delta \in \{1, -1\}$ for which Y_δ is even when U is even, and X_δ is even when V is even. Define

$$a = \gcd(U, X_\delta), \qquad d = \gcd(U, Y_\delta), \qquad S = \frac{X_\delta}{a}, \qquad T = \frac{Y_\delta}{d}. \tag{44}$$

Because $X_\delta Y_\delta = UV$ and $\gcd(X_\delta, Y_\delta) = 1$, every prime power in U occurs wholly in exactly one of X_δ, Y_δ. Therefore $\gcd(U, X_\delta)\gcd(U, Y_\delta) = U$, and equation (44) gives

$$U = ad, \qquad V = ST, \qquad aS = X_\delta, \qquad dT = Y_\delta, \qquad dT - aS = \delta. \tag{45}$$

Figure 3 shows the factor rectangle that equation (45) describes.

The same prime-support separation gives

$$\gcd(a, d) = \gcd(a, T) = \gcd(S, d) = \gcd(S, T) = 1. \tag{46}$$

Our choice of δ makes a, T odd and makes S, d of opposite parity. We claim that

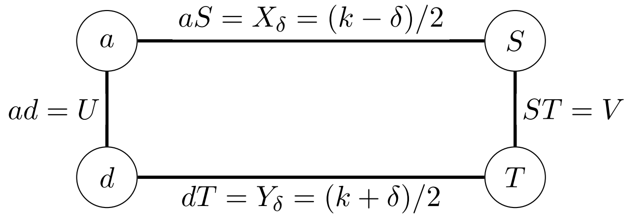

FIGURE 3. The primitive factor rectangle. Coprimality of the consecutive horizontal products allocates every prime power to one corner in each row. The opposite-corner quadratic factors $S^2 + d^2$ and $(T^2 + a^2)/2$ are coprime by equation (47), allowing Lagrange's identity to force two conics.

$$\gcd\left(S^2 + d^2, \frac{T^2 + a^2}{2}\right) = 1. \tag{47}$$

Both factors are odd. If an odd prime p divided them, then none of a, d, S, T would vanish modulo p. Put $r = S/d$ and $s = T/a$ in $\mathbb{F}_p$. The two norm congruences give $r^2 = s^2 = -1$. The relation $dT - aS = \delta$ shows that $r \neq s$, so $s = -r$. It follows that $k = aS + dT = ad(r + s)$ and $W = ST - ad = ad(rs - 1)$ both vanish modulo p, contrary to $\gcd(k, W) = 1$. This proves equation (47).

We therefore have

$$S^2 + d^2 = H^2, \qquad T^2 + a^2 = 2J^2 \tag{48}$$

for coprime positive integers H, J. One direct way to see the square conclusion is Lagrange's identity:

$$(S^2 + d^2)\frac{T^2 + a^2}{2} = \frac{(ad + ST)^2 + (dT - aS)^2}{2} = P_M^2. \tag{49}$$

The two factors on the left are coprime by equation (47), so each is a square.

For the positive Pythagorean pair (S, d), integrality gives $H - S \geqslant 1$ and $H - d \geqslant 1$. Consequently

$$d^2 = (H - S)(H + S) \geqslant 2S + 1, \qquad S^2 = (H - d)(H + d) \geqslant 2d + 1, \tag{50}$$

or equivalently

$$2S \leqslant d^2 - 1, \qquad 2d \leqslant S^2 - 1. \tag{51}$$

Using $k = 2aS + \delta = 2dT - \delta$, the preceding inequalities, and positivity gives

$$k \leqslant a(d^2 - 1) + 1 \leqslant ad^2 \leqslant (ad)^2 = U^2, \tag{52}$$

$$k \leqslant T(S^2 - 1) + 1 \leqslant TS^2 \leqslant (ST)^2 = V^2. \tag{53}$$

These inequalities prove the two factor bounds in equation (39). Finally, the definitions of W show $g = C_M - W = 2U$ on the plus branch and $g = C_M + W = 2V$ on the minus branch. In each case the selected factor is the smaller of U, V, so $g < 2\sqrt{UV} = \sqrt{k^2 - 1} < k$. $\square$

Remark 2.5. The rectangle is reconstructed from the full coefficient-content target, not from a numerical near equality. This is why the lower bound equation (39) is available. The proof of the classification would fail if $g/2$ were replaced by an arbitrary divisor of $(k^2 - 1)/4$.

With the arithmetic dictionary and the primitive rectangle in hand, we now enter the even branch. We first determine which primitive-unit exponents can lead, and then separate those exponents quantitatively from the nearest Pell exponent.

3. Leading unit exponents in the even branch

Assume throughout sections 3 to 5 that an even-exponent target is fixed, and retain j and $M = n - j$ from lemma 2.1. By proposition 2.2, $3 \nmid q$ and M is odd; in particular, lemma 2.4 applies. The first task is to show that every target in the unresolved range has a genuinely negative normalised exponent.

Lemma 3.1 (Ratio bounds)**.** *For $q \geqslant 3$,*

$$\lambda B_q < A_q < 3B_q. \tag{54}$$

Proof. Using $F_{q+1} = F_q + F_{q-1}$, direct expansion gives

$$A_q - \lambda B_q = \sqrt{2}F_q - 2F_{q-1} > 0, \tag{55}$$

because $F_q/F_{q-1} \geqslant 3/2 > \sqrt{2}$ for $q \geqslant 3$. Similarly,

$$3B_q - A_q = 2\big(\sqrt{2}F_{q+1} - 2F_q\big) > 0. \tag{56}$$

Indeed, $F_{q+1}/F_q \geqslant 3/2 > \sqrt{2}$ for $q \geqslant 3$, which proves the last inequality. ☐

Proposition 3.2 (Negative-exponent reduction)**.** *Every even-exponent target with $q \geqslant 7$ has*

$$j = -h, \qquad h \geqslant 3 \text{ odd.} \tag{57}$$

Proof. On the plus branch, equation (40) has positive rational coefficient $W = C_M - g$. Quadratic conjugation gives

$$2W = A_q^2\lambda^{-j} - B_q^2\lambda^j > 0. \tag{58}$$

Thus $\lambda^j < A_q/B_q < 3 < \lambda^2$. Since j is odd, $j \leqslant 1$. On the minus branch the actual rational coefficient is $C_M - g = -W > 0$. Conjugating equation (41) gives

$$2(C_M - g) = B_q^2\lambda^{-j} - A_q^2\lambda^j > 0, \tag{59}$$

and hence

$$\lambda^j < \frac{B_q}{A_q} < \lambda^{-1}. \tag{60}$$

Hence the odd integer j is at most -3 there.

It remains to remove $j = 1$ and $j = -1$ on the plus branch. If $j = 1$, comparison of the $\sqrt{2}$-coefficients in equation (40) gives

$$P_M = F_{q-1}^2 + F_{q+1}^2 = 3F_q^2 + 2(-1)^q. \tag{61}$$

This recurrence equation has exactly

$$(q, M) = (1, 1), (2, 3), (4, 5) \tag{62}$$

among nonnegative q and positive odd M. This follows directly from the complete near-square classification of Alekseyev and Tengely [3, Table 1]. In their notation the Pell sequence is $U_M(2, -1)$. They prove that all Pell values of the forms $3y^2 + 2$ and $3y^2 - 2$ are, respectively,

$$\{2, 5, 29\} \qquad \text{and} \qquad \{1\}. \tag{63}$$

If q is even, equation (61) has the first form in equation (63). The value $2 = P_2$ is removed because M is odd, while $5 = P_3$ and $29 = P_5$ give $F_q = 1$ and $F_q = 3$. Even parity of q leaves $(q, M) = (2, 3), (4, 5)$. If q is odd, the second form in equation (63) gives $P_M = 1$, hence $M = 1$ and $F_q = 1$; odd parity leaves $(q, M) = (1, 1)$. Direct substitution proves the converse.

The nearest plus window is absent at $q = 2$, and $q = 4$ is the low plus target in equation (7). Thus no $j = 1$ target has $q \geqslant 7$.

If $j = -1$, the radical coefficient instead becomes

$$P_M = F_q^2 + 2F_{q+1}^2 + 2F_q F_{q+1} = F_{q+1}^2 + F_{q+2}^2 = F_{2q+3}. \tag{64}$$

The middle expression is also the primitive two-square writing of the Fibonacci-arm Markoff number F_{2q+3}. Thus equation (64) admits the Markoff-arm interpretation developed in section 9. The proof below uses only the Fibonacci identity and Alekseyev's intersection theorem.

Alekseyev's complete Fibonacci–Pell intersection theorem states that the common values of the two sequences are $0, 1, 2, 5$ [2, Theorem 9]. Since $2q + 3 \geqslant 17$ for $q \geqslant 7$, equation (64) is impossible. All remaining exponents therefore have the form (57). ☐

Remark 3.3 (Independent verification of the near-square step)**.** The published theorem [3] is the completeness input; this boundary classification is not new here. As an independent audit, the negative-Pell identity converts equation (61), with $y = F_q$ and $s = (-1)^q$, into

$$C_M^2 = 18y^4 + 24sy^2 + 7. \tag{65}$$

After $y = z + 1$, two complete Magma V2.29-9 integral-point computations give exactly the points corresponding to equation (62). The supplement contains both inputs, complete transcripts, and independent arithmetic verifiers. These computations corroborate the published theorem and are not used as an unrecorded bounded search.

Thus every even-exponent target in the unresolved range has primitive unit direction λ^{-h} with $h \geqslant 3$ odd. The qualitative leading-unit ambiguity has disappeared; the next task is quantitative, namely to separate the negative exponent h from the nearest Pell exponent n.

4. Height separation in the even branch

Continue under the standing hypothesis of an even-exponent target with $q \geqslant 7$. By proposition 3.2,

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^{-h}, \qquad h \geqslant 3 \text{ odd}. \tag{66}$$

We first turn the reconstructed rectangle into a separation between h and n.

Lemma 4.1 (Divisor-height separation)**.** *Every target in equation (66) satisfies*

$$\begin{aligned} \sigma = 1: &\quad h \leqslant n - 5, \\ \sigma = -1: &\quad h \leqslant n - 3. \end{aligned} \tag{67}$$

In particular, $h < n$.

Proof. Put

$$L = f^2 + 2x^2. \tag{68}$$

Since $\lambda^{-h} = -C_h + P_h\sqrt{2}$ for odd h, coefficient comparison in equation (66) gives

$$gC_h = L - C_n, \qquad gP_h = P_n - 2\sigma f x. \tag{69}$$

Let $t = g/2$. By lemma 2.4, $t \geqslant \sqrt{k}$. For every positive odd h,

$$C_h = \frac{\lambda^h - \lambda^{-h}}{2} > \frac{\lambda^h}{3}. \tag{70}$$

On the plus branch, $L < A_q^2 < \lambda^n$. Hence

$$t = \frac{L - C_n}{2C_h} < \frac{3}{2}\lambda^{n-h}. \tag{71}$$

On the minus branch, equation (54) gives

$$L = \frac{A_q^2 + B_q^2}{2} < 5B_q^2 < 5\lambda^n, \tag{72}$$

and therefore $t < (15/2)\lambda^{n-h}$. Combining these inequalities with $t \geqslant \sqrt{k}$ yields

$$\lambda^{2(n-h)} > \begin{cases} 4k/9, & \sigma = 1, \\ 4k/225, & \sigma = -1. \end{cases} \tag{73}$$

The sequence R_q is increasing for $q \geqslant 3$: with $f = F_q$ and $x = F_{q+1} < 2f$,

$$R_{q+1} - R_q = 3f^2 + 4fx - x^2 > 0. \tag{74}$$

Hence $k \geqslant R_7 = 713$, and $n - h$ is odd. Direct comparison in equation (73) excludes $n-h \leqslant 3$ in the plus case and $n-h \leqslant 1$ in the minus case. This proves equation (67). ☐

We shall use the following explicit specialisation of Matveev's theorem [16, Corollary 2.3]. For a positive algebraic number α, write $h_{\mathrm{abs}}(\alpha)$ for its absolute logarithmic Weil height.

Lemma 4.2 (Matveev, three positive real logarithms). *Let K be a real number field of degree D, let $\alpha_1, \alpha_2, \alpha_3 \in K$ be positive, and let*

$$\Lambda = b_1 \log\alpha_1 + b_2 \log\alpha_2 + b_3 \log\alpha_3 \neq 0 \tag{75}$$

with $b_i \in \mathbb{Z}$. If

$$A_i \geqslant \max\{Dh_{\mathrm{abs}}(\alpha_i), |\log\alpha_i|, 0.16\} \tag{76}$$

and $B \geqslant \max_i |b_i|$, then

$$\log|\Lambda| > -1.4 \cdot 30^6 3^{9/2} D^2(1 + \log D)A_1A_2A_3(1 + \log B). \tag{77}$$

The first application reduces the a priori linear range for h to a logarithmic range.

Proposition 4.3 (Logarithmic bound for the negative exponent). *Every target with $q \geqslant 7$ satisfies*

$$h < 2.4 \cdot 10^{14}(1 + \log(2q)) \tag{78}$$

and

$$n < 2q. \tag{79}$$

Proof. Put

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \epsilon = (-1)^q, \qquad u = \sigma + \sqrt{2}\varphi, \qquad v = -\sigma + \sqrt{2}\varphi^{-1}. \tag{80}$$

Binet's formula gives

$$D_{q,\sigma} = \frac{u\varphi^q + \epsilon v \varphi^{-q}}{\sqrt{5}}. \tag{81}$$

Set

$$\delta = \epsilon \frac{v}{u}\varphi^{-2q}, \qquad A_0 = \frac{u^2}{5}\varphi^{2q}. \tag{82}$$

Then $D_{q,\sigma}^2 = A_0(1+\delta)^2$. The elementary bounds $1 < u < 4$ and $|v/u| < 2$ give

$$|\delta| < 2\left(\frac{2}{3}\right)^{14} < \frac{1}{100}. \tag{83}$$

By lemma 4.1 and the nearest window,

$$\lambda^{h+3} < \frac{32}{5}\left(\frac{101}{100}\right)^2 \varphi^{2q} < 7\varphi^{2q}. \tag{84}$$

Since $\lambda > 12/5$, this strengthens equation (83) to

$$|\delta| < \frac{1750}{1728}\lambda^{-h}. \tag{85}$$

By lemma 2.4, $g < k = A_q B_q$. Therefore $g/D_{q,\sigma}^2 < 3$ for either sign, by equation (54). Define

$$\Lambda_0 = n\log\lambda - 2q\log\varphi - \log\left(\frac{u^2}{5}\right). \tag{86}$$

After dividing equation (66) by A_0, the estimates equations (83) and (85) give

$$|e^{\Lambda_0} - 1| < \frac{128}{25}\lambda^{-h} < \frac{10}{27}. \tag{87}$$

The elementary logarithm estimate on this interval yields

$$0 < |\Lambda_0| < 9\lambda^{-h}. \tag{88}$$

The nonvanishing is exact. In $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$ one has $N_{K/\mathbb{Q}}(u) = -1$ and $N_{K/\mathbb{Q}}(\lambda) = N_{K/\mathbb{Q}}(\varphi) = 1$. Equality in equation (86) would give $1 = 5^{-4}$ after taking full norms.

We next verify equation (79). From equation (81), $D_{q,\sigma} < 4\varphi^q$. Also $\lambda/\varphi > 7/5$ and $(7/5)^{14} > 32$. Thus, for $q \geqslant 7$,

$$\lambda^{2q} > 32\varphi^{2q} > 2D_{q,\sigma}^2 > \lambda^n. \tag{89}$$

For lemma 4.2, the field degree is 4 and admissible parameters for equation (86) are

$$A_1 = 2\log\lambda, \qquad A_2 = 2\log\varphi, \qquad A_3 = 20, \qquad B = 2q. \tag{90}$$

Indeed,

$$h_{\text{abs}}(u^2/5) \leqslant 3\log 2 + \log\varphi + \log 5 < 5, \tag{91}$$

and $1/5 < u^2/5 < 16/5$. The constant before A_3 satisfies the exact outward inequality

$$1.4 \cdot 30^6 3^{9/2} 4^2 (1+\log 4)(2\log\lambda)(2\log\varphi) < 9.275 \cdot 10^{12}. \tag{92}$$

Writing $H = 1 + \log(2q)$, comparison with equation (88) gives

$$h \log \lambda < 1.855 \cdot 10^{14} H + \log 9. \tag{93}$$

Finally $\log \lambda > 4/5$. Rounding only upward proves equation (78). ☐

The fixed-coefficient form has therefore done two jobs: it makes h logarithmic in q and keeps n below $2q$. It does not yet bound q. The next section supplies that missing absolute bound by allowing the third logarithmic coefficient to depend on the now-controlled exponent h.

5. Effective classification of the even branch

We now bound q, reduce both exponents by exact rational intervals, and finish the even-exponent branch of theorem 1.1. Until the terminal enumeration, we retain the standing hypothesis that an even-exponent target exists with $q \geqslant 7$.

5.1. **A moving-coefficient logarithmic form.** Quadratic conjugation of equation (66), followed by elimination of g, gives the exact trace identity

$$\lambda^n + \lambda^{-n-2h} = D_{q,\sigma}^2 + \overline{D}_{q,\sigma}^2 \lambda^{-2h}. \tag{94}$$

Retain $\epsilon = (-1)^q$ and put

$$\begin{aligned} u_\sigma &= \sigma + \sqrt{2}\varphi, \quad u'_\sigma = -\sigma + \sqrt{2}\varphi^{-1}, \\ v_\sigma &= \sigma - \sqrt{2}\varphi, \quad v'_\sigma = -\sigma - \sqrt{2}\varphi^{-1}. \end{aligned} \tag{95}$$

Binet's formula becomes

$$\begin{aligned} D_{q,\sigma} &= \frac{u_\sigma \varphi^q + \epsilon u'_\sigma \varphi^{-q}}{\sqrt{5}}, \\ \overline{D}_{q,\sigma} &= \frac{v_\sigma \varphi^q + \epsilon v'_\sigma \varphi^{-q}}{\sqrt{5}}. \end{aligned} \tag{96}$$

The key is to retain both growing φ^{2q} terms inside one coefficient:

$$\mathcal{A}_{\sigma,h} = \frac{u_\sigma^2 + v_\sigma^2 \lambda^{-2h}}{5}. \tag{97}$$

Substitution in equation (94) gives

$$\lambda^n = \mathcal{A}_{\sigma,h} \varphi^{2q} + \mathcal{R}_{\sigma,q,h}, \tag{98}$$

where

$$\begin{aligned} \mathcal{R}_{\sigma,q,h} = \frac{2\epsilon}{5} &\Big((1 - \sigma\sqrt{2}) + (1 + \sigma\sqrt{2})\lambda^{-2h} \Big) \\ &+ \frac{\varphi^{-2q}}{5} \Big((u'_\sigma)^2 + (v'_\sigma)^2 \lambda^{-2h} \Big) - \lambda^{-n-2h}. \end{aligned} \tag{99}$$

Here we used $u_\sigma u'_\sigma = 1 - \sigma\sqrt{2}$ and $v_\sigma v'_\sigma = 1 + \sigma\sqrt{2}$. Direct coefficient estimates give

$$|\mathcal{R}_{\sigma,q,h}| < 4, \qquad \frac{1}{5} < \mathcal{A}_{\sigma,h} < 7. \tag{100}$$

The proof is at $q \geqslant 7$, so $\varphi^{2q} \geqslant \varphi^{14}$. Since $\varphi > 8/5$ one has

$$\varphi^{14} = 377\varphi + 233 > 842, \tag{101}$$

which is more than the value 420 needed to pass from $|e^{\Lambda_1} - 1| < 20\varphi^{-2q}$ to the displayed logarithmic bound.[1] Normalising equation (98) and using equation (100) yields the logarithmic form

$$\Lambda_1 = n \log \lambda - 2q \log \varphi - \log \mathcal{A}_{\sigma,h} \tag{102}$$

with

$$0 < |\Lambda_1| < 21\varphi^{-2q}. \tag{103}$$

The strict nonvanishing and the height of the moving coefficient must be controlled uniformly. From $\lambda^h = C_h + P_h\sqrt{2}$ one obtains

$$\mathcal{A}_{\sigma,h} = \lambda^{-h}\frac{2\sqrt{2}}{5}\left((4+\sqrt{5})P_h + 2\sigma\varphi C_h\right). \tag{104}$$

Taking the full norm in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ gives

$$N_{K/\mathbb{Q}}(\mathcal{A}_{\sigma,h}) = \frac{64}{625}\left(4 + 3P_h^2 - 2\sigma C_h P_h\right)^2. \tag{105}$$

If $\Lambda_1 = 0$, full norms would force $64I^2 = 625$ for an integer I, which is impossible. We now record the height calculation. The inequalities $h_{\text{abs}}(\alpha\beta) \leqslant h_{\text{abs}}(\alpha) + h_{\text{abs}}(\beta)$ and $h_{\text{abs}}(\alpha + \beta) \leqslant h_{\text{abs}}(\alpha) + h_{\text{abs}}(\beta) + \log 2$ give

$$h_{\text{abs}}(u_\sigma),\ h_{\text{abs}}(v_\sigma) \leqslant \log 2 + \frac{\log 2}{2} + \frac{\log \varphi}{2} < \frac{3}{2}. \tag{106}$$

Since $h_{\text{abs}}(\lambda) = \frac{1}{2}\log\lambda$, the definition equation (97) yields

$$\begin{aligned} h_{\text{abs}}(\mathcal{A}_{\sigma,h}) &\leqslant 2h_{\text{abs}}(u_\sigma) + 2h_{\text{abs}}(v_\sigma) + 2h\, h_{\text{abs}}(\lambda) + \log 10 \\ &< h + 9 < h + 11. \end{aligned} \tag{107}$$

Together with equation (100), this proves

$$h_{\text{abs}}(\mathcal{A}_{\sigma,h}) < h + 11, \qquad |\log \mathcal{A}_{\sigma,h}| < 2. \tag{108}$$

Thus lemma 4.2 applies to equation (102) with

$$A_1 = 2\log\lambda, \qquad A_2 = 2\log\varphi, \qquad A_3 = 4h + 44, \qquad B = 2q. \tag{109}$$

Writing again $H = 1 + \log(2q)$ and using equation (92), comparison with equation (103) gives

$$2q\log\varphi < 9.275 \cdot 10^{12}(4h + 44)H + \log 21. \tag{110}$$

By equation (78) and $\log\varphi > 2/5$,

$$q < \frac{5}{4}\left[9.275 \cdot 10^{12}\left(9.6 \cdot 10^{14}H + 44\right)H + 4\right]. \tag{111}$$

At $q_0 = 10^{32}$, one has $H(q_0) < 79$ and the right-hand side of equation (111) is less than $7 \cdot 10^{31}$. After division by q, every term on the right is decreasing for $q \geqslant q_0$. Consequently

$$q < 10^{32}. \tag{112}$$

[1] The coefficients here are not accidental. The Christoffel word attached to the Markoff number $F_{15} = 610$ has exactly $F_{14} = 377$ letters a and $F_{13} = 233$ letters b; see section 9.

The moving coefficient equation (97) is the central mechanism of the proof. Its error decays with q, while its height grows only linearly with h, which is already logarithmic in q.

5.2. **An absolute-value reduction lemma.** We use a symmetric form of the Baker–Davenport reduction, closely related to the lemma of Dujella and Pethő [7]. We include a short proof to avoid a hidden sign condition.

Lemma 5.1 (Absolute-value reduction)**.** *Let $\gamma, \mu \in \mathbb{R}$, let $p \in \mathbb{Z}$, let $Q \in \mathbb{Z}_{>0}$, and let $N_0 \geqslant 1$. Suppose that $|Q\gamma - p| < 1/2$ and that integers a, b satisfy*

$$0 < |a\gamma - b + \mu| < \mathcal{C}\rho^{-w}, \qquad 1 \leqslant a \leqslant N_0, \tag{113}$$

where $\mathcal{C} > 0$, $\rho > 1$, and $w > 0$ are real. Put

$$\kappa = \|Q\mu\| - N_0|Q\gamma - p|, \tag{114}$$

where $\|x\|$ is the distance from x to the nearest integer. If $\kappa > 0$, then

$$w < \frac{\log(\mathcal{C}Q/\kappa)}{\log \rho}. \tag{115}$$

The symbols $\mathcal{C}$, ρ, N_0 and κ are local to this lemma and to its four applications below: two in section 5 and two in section 6.

Proof. Since $ap - bQ \in \mathbb{Z}$, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ gives

$$\begin{aligned} \|Q\mu\| &\leqslant \|Q\mu + a(Q\gamma - p)\| + a|Q\gamma - p| \\ &\leqslant Q|a\gamma - b + \mu| + N_0|Q\gamma - p|. \end{aligned} \tag{116}$$

The strict hypothesis therefore gives $\kappa < \mathcal{C}Q\rho^{-w}$, which is equivalent to equation (115). ☐

The point of this symmetric form is that no sign of the logarithmic form is assumed. We apply it first to the fixed shift, reducing h or q, and then to the moving shift, which supplies the absolute bound for q.

5.3. **Certified reductions of the two exponents.** Recall the phase constant γ of equation (33) and set

$$\gamma = \frac{\log\varphi}{\log\lambda}, \qquad \mu_\sigma = \frac{\log(u_\sigma^2/5)}{\log\lambda}. \tag{117}$$

The same constant governs the two reductions and the density theorem, and it has an intrinsic meaning: it is the exchange rate between the two extreme arms of the Markoff tree, in the sense made precise in section 9.

Put

$$\delta_\sigma = \epsilon\frac{u'_\sigma}{u_\sigma}\varphi^{-2q}, \qquad \widetilde{\delta}_\sigma = \epsilon\frac{v'_\sigma}{v_\sigma}\varphi^{-2q}. \tag{118}$$

Dividing the Binet expansion of equation (94) by $(u_\sigma^2/5)\varphi^{2q}$ gives the exact identity

$$\begin{aligned} e^{\Lambda_0} - 1 = 2\delta_\sigma + \delta_\sigma^2 + \frac{v_\sigma^2}{u_\sigma^2}(1+\widetilde{\delta}_\sigma)^2\lambda^{-2h} \\ - \frac{5}{u_\sigma^2}\lambda^{-n-2h}\varphi^{-2q}. \end{aligned} \tag{119}$$

For both signs, direct comparison gives

$$1 < u_\sigma < 4, \qquad |\frac{u'_\sigma}{u_\sigma}| < \frac{3}{2}, \qquad |\frac{v_\sigma}{u_\sigma}| < 3, \qquad |\frac{v'_\sigma}{v_\sigma}| < \frac{3}{2}. \tag{120}$$

Since $q \geqslant 7$, equation (101) gives $\varphi^{2q} > 842 > 150$, so both delta terms, which are bounded by $(3/2)\varphi^{-2q}$, have absolute value below $1/100$. The three terms in equation (119) are therefore bounded, respectively, by $4\varphi^{-2q}$, $10\lambda^{-2h}$, and $5\varphi^{-2q}$. Moreover, $h \geqslant 3$ and $\lambda^6 > 190$, whence

$$|e^{\Lambda_0} - 1| < 10\left(\lambda^{-2h} + \varphi^{-2q}\right) < \frac{1}{10}. \tag{121}$$

The inequality $|\log(1+x)| \leqslant |x|/(1-|x|)$ for $|x| < 1$ now gives the sharper estimate

$$0 < |\Lambda_0| < 12\left(\lambda^{-2h} + \varphi^{-2q}\right), \tag{122}$$

where Λ_0 is defined in equation (86). Put $w = \min\{h, q\gamma\}$. Since $\log\lambda > 4/5$, equation (122) implies

$$0 < |2q\gamma - n + \mu_\sigma| < 30(\lambda^2)^{-w}. \tag{123}$$

Take

$$N_0 = 2 \cdot 10^{32} \tag{124}$$

and the rational number

$$\frac{p}{Q} = \frac{258576277494354008456674440956683 3}{47360079147084818102211862783095 84}. \tag{125}$$

Exact rational intervals for square roots and logarithms certify

$$\gcd(p, Q) = 1, \qquad Q > 6N_0, \qquad 2Q|Q\gamma - p| < 1, \qquad |Q\gamma - p| < \frac{26}{10^{36}}. \tag{126}$$

The first three conditions are the hypotheses of lemma 5.1; the fourth is the quantitative enclosure that the margins below actually consume. It is strictly stronger than the Legendre bound $|Q\gamma - p| < 1/(2Q)$, which by itself would leave $N_0|Q\gamma - p| < 22/1000$ and would not suffice. With the displayed enclosure,

$$N_0|Q\gamma - p| < \frac{52}{10^4}. \tag{127}$$

Remark 5.2 (Where p/Q comes from)**.** The integers of equation (125) are not produced by the proof; they are supplied to it. They were found by expanding γ in continued fractions and taking the first convergent whose denominator exceeds $6N_0$. The restriction to convergents is not a matter of taste. The reduction lemma itself asks only for $|Q\gamma - p| < 1/2$; what forces the search into the convergents is the approximation quality the margins consume, which lies far beyond the Legendre threshold $|Q\gamma - p| < 1/(2Q^2)$, and by Legendre's theorem the only rationals that good are the convergents of γ. The size condition $Q > 6N_0$ then keeps the parasitic term $N_0|Q\gamma - p|$ well below the value $1/2$ that $\|Q\mu_\sigma\|$ can never exceed. None of this search enters the proof. Once the two integers are written down they are treated as given data, and equation (126) re-establishes by exact rational arithmetic every property of them that is used. The search is heuristic; the verification is exact.

Only the certified error is used below. For both signs, the same interval certificate gives

$$\|Q\mu_\sigma\| > \frac{82}{1000}, \qquad \kappa_\sigma = \|Q\mu_\sigma\| - N_0|Q\gamma - p| > \frac{7}{100}, \tag{128}$$

where the second inequality follows from the first together with equation (127). The same certificate gives

$$\kappa_\sigma(\lambda^2)^{48} > 30Q. \tag{129}$$

Apply lemma 5.1 with

$$a = 2q, \quad b = n, \quad \mu = \mu_\sigma, \quad \mathcal{C} = 30, \quad \rho = \lambda^2, \quad w = \min\{h, q\gamma\}, \quad N_0 = 2 \cdot 10^{32}. \tag{130}$$

Equations (126), (128) and (129) then prove $\min\{h, q\gamma\} < 48$. Since $\gamma > 1/2$, either $h < 48$, or $q < 96$ and then

$$h < n < 2q < 192. \tag{131}$$

Thus every target with $q \geqslant 7$ satisfies

$$3 \leqslant h < 192, \qquad h \text{ odd}. \tag{132}$$

For each of the 190 pairs

$$\sigma \in \{1, -1\}, \qquad h \in \{3, 5, \dots, 191\}, \tag{133}$$

put

$$\nu_{\sigma,h} = \frac{\log \mathcal{A}_{\sigma,h}}{\log \lambda}. \tag{134}$$

Equation (103) gives

$$0 < |2q\gamma - n + \nu_{\sigma,h}| < 27(\varphi^2)^{-q}. \tag{135}$$

The same p/Q satisfies, for every one of the 190 shifts,

$$\|Q\nu_{\sigma,h}\| > \frac{143}{10^4}, \qquad \kappa_{\sigma,h} = \|Q\nu_{\sigma,h}\| - N_0|Q\gamma - p| > \frac{9}{1000} \tag{136}$$

where again the second inequality follows from the first and equation (127), and

$$\kappa_{\sigma,h}(\varphi^2)^{90} > 27Q. \tag{137}$$

The weakest certified interval occurs at $(\sigma, h) = (1, 31)$, where $\|Q\nu_{\sigma,h}\| = 0.01436\ldots$; it remains strictly above the displayed margin. This is also the reason the sharper enclosure in equation (126) is displayed: the Legendre bound alone would make $\kappa_{1,31}$ negative, and the reduction lemma would not apply.

Apply lemma 5.1 for each pair with

$$a = 2q, \quad b = n, \quad \mu = \nu_{\sigma,h}, \quad \mathcal{C} = 27, \quad \rho = \varphi^2, \quad w = q, \quad N_0 = 2 \cdot 10^{32}. \tag{138}$$

Equations (126), (136) and (137) prove

$$q < 90. \tag{139}$$

The interval proof uses only rational arithmetic. For a positive rational x, logarithms are enclosed using

$$\log x = 2 \sum_{r=0}^{N-1} \frac{z^{2r+1}}{2r+1} + R_N, \qquad z = \frac{x-1}{x+1}, \tag{140}$$

with the geometric tail bound

$$|R_N| \leqslant \frac{2|z|^{2N+1}}{(2N+1)(1-z^2)}. \tag{141}$$

Square roots are bounded between consecutive rational grid points. Hence no binary floating-point comparison enters equations (126), (128), (129), (136) and (137).

At this point the unbounded even-exponent problem has become finite: every hypothetical target has $q < 90$. What remains is an exact recurrence calculation, not a further asymptotic estimate.

5.4. **Exact terminal enumeration.** The remaining lower boundary $q = 1$ is direct:

$$D_{1,1} = \lambda, \qquad D_{1,-1} = \lambda^{-1}. \tag{142}$$

For the plus sign, the first integral power strictly above $D_{1,1}^2 = \lambda^2$ is $\lambda^3 > 2\lambda^2$. For the minus sign, every positive integral power is at least $\lambda > 2\lambda^{-2} = 2D_{1,-1}^2$. Thus neither sign has a strict candidate at $q = 1$.

It remains to test $2 \leqslant q < 90$, $3 \nmid q$, and both signs. For each row, exact recurrence arithmetic constructs $D_{q,\sigma}^2$ and the first even Pell unit above it. The algorithm then checks the strict upper window, computes the two coefficient gcd, and tests the exact norm identity. Signs of $a + b\sqrt{2}$ are decided over the integers from a, b, and $a^2 - 2b^2$; no numerical approximation is used.

The complete output consists of

$$q = 2, \sigma = -1: \quad \lambda^2 - (-F_2 + F_3\sqrt{2})^2 = 6\lambda^{-1}, \tag{143}$$

$$q = 4, \sigma = 1: \quad \lambda^6 - (F_4 + F_5\sqrt{2})^2 = 40\lambda. \tag{144}$$

No other row passes. In particular, the range $7 \leqslant q < 90$ is empty; together with equation (139), this proves the even-exponent branch of theorem 1.1. The odd branch cannot use the primitive rectangle or the inequality $h < n$; the next section closes it by reversing the order of the fixed and moving logarithmic-form estimates.

6. The odd-exponent branch

The preceding sections classified the even-exponent branch. We now return to the intrinsic setup of section 2 and suppose that the exponent n is odd. The parity selector forces $3 \mid q$, so this branch is disjoint from the primitive-rectangle argument. Its content has a different factorisation, and in particular no analogue of the divisor-height separation lemma 4.1 is available. The proof closes after reversing the order of the two logarithmic-form estimates.

Proposition 6.1 (Negative unit exponent in the odd branch)**.** *Suppose that* $q \geqslant 1$, $\sigma \in \{1, -1\}$, *and a positive odd integer* n *satisfy equations (23) and (24). Then* $3 \mid q$, *and the unit normalisation has the form*

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^{-h}, \qquad h \geqslant 3 \text{ odd}. \tag{145}$$

Proof. By proposition 2.2, $3 \mid q$ and hence $q \geqslant 3$. Put

$$A_+ = F_q + F_{q+1}\sqrt{2}, \qquad A_- = -F_q + F_{q+1}\sqrt{2}. \tag{146}$$

Both are positive. By lemma 2.1, there is a unique odd integer j such that

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^j. \tag{147}$$

Since $\overline{D_{q,\sigma}} = -D_{q,-\sigma}$ and both n and j are odd, conjugation gives

$$\lambda^{-n} + D_{q,-\sigma}^2 = g\lambda^{-j}. \tag{148}$$

Division of equation (147) by equation (148), followed by the strict upper window, yields

$$\lambda^{2j} < \frac{D_{q,\sigma}^2}{D_{q,-\sigma}^2}. \tag{149}$$

The ratio bounds lemma 3.1 say $\lambda A_- < A_+ < 3A_-$. Thus $j \leqslant 1$ for $\sigma = 1$, whereas $j \leqslant -3$ for $\sigma = -1$.

It remains to remove $j = 1$ and $j = -1$ on the plus branch. The following coefficient argument in fact excludes those values for either sign. Set

$$H_+ = F_{q+1}^2 + F_{q+2}^2 = F_{2q+3}, \qquad H_- = F_{q+1}^2 + F_{q-1}^2 = 3F_q^2 + 2(-1)^q. \tag{150}$$

If $j = 1$, multiply equation (147) by λ^{-1} and compare the $\sqrt{2}$-coefficients; if $j = -1$, multiply it by λ instead. The resulting equations are

$$P_{n-1} = H_{-\sigma} \quad \text{or} \quad P_{n+1} = H_\sigma, \tag{151}$$

respectively. Alekseyev's complete Fibonacci–Pell intersection theorem states that the common values of the two sequences are $0, 1, 2, 5$ [2, Theorem 9]. Hence $H_+ = F_{2q+3} \geqslant F_9 = 34$ cannot occur. Alekseyev and Tengely's complete near-square table gives

$$3y^2 + 2 \in \{2, 5, 29\}, \qquad 3y^2 - 2 \in \{1\} \tag{152}$$

whenever the displayed number is a Pell value [3, Table 1]. If q is odd, then $H_- = 3F_q^2 - 2 \geqslant 10$; if q is even, then $H_- = 3F_q^2 + 2 \geqslant 194$. Thus neither alternative in equation (151) is possible. Every remaining j is at most -3, proving equation (145) with $j = -h$. ☐

Lemma 6.2 (Odd trace and content factorisation)**.** *Under the hypotheses of proposition 6.1, put*

$$f = F_q, \qquad x = F_{q+1}, \qquad L = f^2 + 2x^2, \qquad R_q = 2x^2 - f^2, \qquad M = n + h. \tag{153}$$

Then

$$\lambda^n - \lambda^{-n-2h} = D_{q,\sigma}^2 + D_{q,-\sigma}^2 \lambda^{-2h}, \tag{154}$$

and

$$R_q^2 + 1 = g(2C_M - g), \qquad 1 \leqslant g < R_q, \qquad g \mid R_q^2 + 1. \tag{155}$$

In particular, every prime divisor of g is congruent to 1 modulo 4.

Proof. Conjugating equation (145) gives

$$\lambda^{-n} + D_{q,-\sigma}^2 = g\lambda^h. \tag{156}$$

Eliminating g between this equation and equation (145) proves equation (154). Since $\lambda^{-h} = -C_h + P_h\sqrt{2}$, coefficient comparison gives

$$L - C_n = gC_h, \qquad P_n - 2\sigma f x = gP_h. \tag{157}$$

Taking norms in equation (145), and using that $M = n + h$ is even, gives

$$R_q^2 + 1 = g(2C_M - g). \tag{158}$$

The Pell addition formulas and equation (157) also give

$$C_M - g = LC_h + 4\sigma f x P_h. \tag{159}$$

This is positive for $\sigma = 1$. For $\sigma = -1$, use

$$L - 3fx = (x - f)(2x - f) > 0, \qquad \frac{C_h}{P_h} = \sqrt{2 - \frac{1}{P_h^2}} \geqslant \frac{7}{5} > \frac{4}{3} \qquad (h \geqslant 3 \text{ odd}) \tag{160}$$

to obtain positivity again. The middle identity is equation (17) at odd h, and $P_h \geqslant P_3 = 5$ gives the stated bound; the restriction to $h \geqslant 3$ is needed, since $3C_1 = 3 < 4 = 4P_1$. Hence $g < C_M$, so $g^2 < R_q^2 + 1$. Here g is odd by the coefficient parity in equation (27), whereas R_q is even because $3 \mid q$. Therefore $g < R_q$, and the divisibility assertion follows from the first factorisation in equation (155). Finally, if a prime p divides g, then p is odd and $R_q^2 \equiv -1 \pmod{p}$; hence $p \equiv 1 \pmod{4}$. ☐

The factorisation in lemma 6.2 explains why the even proof cannot simply be quoted. It gives no lower bound for g and no inequality $h < n$. We therefore use the fixed logarithmic form first, control $\min\{h, q\gamma\}$, and only then invoke the moving form.

Theorem 6.3 (Odd-exponent exclusion). *For $q \geqslant 1$ and $\sigma \in \{1, -1\}$, no positive odd integer n can satisfy both equations (23) and (24).*

Proof. Suppose otherwise. By proposition 6.1, $3 \mid q$ and equation (145) holds. We first treat $q \geqslant 7$; the indices $q = 3, 6$ will be included in the exact terminal enumeration.

Retain φ, $\epsilon = (-1)^q$, and the four Binet coefficients from equations (95) and (96), and put

$$\gamma = \frac{\log \varphi}{\log \lambda}, \qquad H = 1 + \log(2q). \tag{161}$$

The window and the elementary bound $D_{q,\sigma} < 4\varphi^q$ give

$$\lambda^n < 32\varphi^{2q} < \lambda^{2q}, \qquad n < 2q. \tag{162}$$

Indeed, $\lambda/\varphi > 7/5$ and $(7/5)^{14} > 32$.
We begin with the fixed form

$$\Lambda_0 = n \log \lambda - 2q \log \varphi - \log\left(\frac{u_\sigma^2}{5}\right). \tag{163}$$

Define

$$\delta_\sigma = \epsilon \frac{u'_\sigma}{u_\sigma} \varphi^{-2q}, \qquad \widetilde{\delta}_\sigma = \epsilon \frac{v'_\sigma}{v_\sigma} \varphi^{-2q}. \tag{164}$$

Expanding equation (154) gives the exact identity

$$e^{\Lambda_0}-1=2\delta_\sigma+\delta_\sigma^2+\frac{v_\sigma^2}{u_\sigma^2}(1+\widetilde{\delta}_\sigma)^2\lambda^{-2h}\\ +\frac{5}{u_\sigma^2}\lambda^{-n-2h}\varphi^{-2q}. \tag{165}$$

The final sign is positive; this is the precise change from equation (119). The coefficient estimates used there are insensitive to this sign and give

$$0<|\Lambda_0|<12\left(\lambda^{-2h}+\varphi^{-2q}\right). \tag{166}$$

Nonvanishing is exact: if $\Lambda_0=0$, full norms in $\mathbb{Q}(\sqrt{2},\sqrt{5})$ would give $1=5^{-4}$. By lemma 4.2 and equations (92) and (162), the same explicit height parameters as in section 4 yield

$$\log|\Lambda_0|>-1.855\cdot 10^{14}H. \tag{167}$$

Put

$$w=\min\{h,q\gamma\}. \tag{168}$$

The right-hand side of equation (166) is at most $24\lambda^{-2w}$. Since $\log\lambda>4/5$ and $\log 24<4$, comparison with equation (167) gives

$$w<1.16\cdot 10^{14}H. \tag{169}$$

We next use the moving coefficient $\mathcal{A}_{\sigma,h}$ from equation (97). The form

$$\Lambda_1=n\log\lambda-2q\log\varphi-\log\mathcal{A}_{\sigma,h} \tag{170}$$

has the exact remainder

$$\mathcal{R}^{\mathrm{odd}}_{\sigma,q,h}=\frac{2\epsilon}{5}\left((1-\sigma\sqrt{2})+(1+\sigma\sqrt{2})\lambda^{-2h}\right)\\ +\frac{\varphi^{-2q}}{5}\left((u'_\sigma)^2+(v'_\sigma)^2\lambda^{-2h}\right)+\lambda^{-n-2h}. \tag{171}$$

Thus

$$\lambda^n=\mathcal{A}_{\sigma,h}\varphi^{2q}+\mathcal{R}^{\mathrm{odd}}_{\sigma,q,h}. \tag{172}$$

Only the sign of the last term differs from equation (99). The same absolute estimates, full norm equation (105), and height bound equation (108) therefore give

$$0<|\Lambda_1|<21\varphi^{-2q} \tag{173}$$

and, by lemma 4.2 and equation (92) with the moving height parameters equation (109), whose bound $B=2q$ is admissible here by equation (162),

$$2q\log\varphi<9.275\cdot 10^{12}(4h+44)H+\log 21. \tag{174}$$

If $q\geqslant 10^{32}$, then $\gamma>1/2$ and the monotonicity of q/H show that $q\gamma$ exceeds the right-hand side of equation (169). Hence $w=h$. At $q_0=10^{32}$ one has $H(q_0)<79$. Substitution of $h<1.16\cdot 10^{14}H$ into equation (174), using $\log\varphi>2/5$, gives at q_0

$$q<\frac{5}{4}\left[9.275\cdot 10^{12}\left(4(1.16\cdot 10^{14})H+44\right)H+4\right]<3.36\cdot 10^{31}. \tag{175}$$

After division by q, the terms H^2/q, H/q, and $1/q$ decrease for $q \geqslant q_0$, so the contradiction persists. Consequently

$$q < 10^{32}. \tag{176}$$

It remains to apply the exact reductions already used in section 5. Put

$$\mu_\sigma = \frac{\log(u_\sigma^2/5)}{\log \lambda}. \tag{177}$$

Dividing equation (166) by $\log \lambda$ gives

$$0 < |2q\gamma - n + \mu_\sigma| < 30(\lambda^2)^{-w}, \qquad 2q < 2 \cdot 10^{32}. \tag{178}$$

The common rational approximation and the two fixed-shift certificates equations (125), (126), (128) and (129) depend only on γ, μ_σ, and λ, not on the parity of n. The absolute-value reduction in lemma 5.1, applied with

$$a = 2q, \quad b = n, \quad \mu = \mu_\sigma, \quad \mathcal{C} = 30, \quad \rho = \lambda^2, \quad w = \min\{h, q\gamma\}, \quad N_0 = 2 \cdot 10^{32}, \tag{179}$$

therefore proves

$$w < 48. \tag{180}$$

If $q \geqslant 96$, then $q\gamma > 48$, so $h = w < 48$. Put

$$\nu_{\sigma,h} = \frac{\log \mathcal{A}_{\sigma,h}}{\log \lambda}. \tag{181}$$

Equation (173) gives

$$0 < |2q\gamma - n + \nu_{\sigma,h}| < 27(\varphi^2)^{-q}. \tag{182}$$

The moving certificates equations (136) and (137) cover both signs and every odd $3 \leqslant h < 192$, hence in particular the present range $3 \leqslant h < 48$. A second application of lemma 5.1, now with $a = 2q$, $b = n$, $\mu = \nu_{\sigma,h}$, $\mathcal{C} = 27$, $\rho = \varphi^2$, $w = q$ and $N_0 = 2 \cdot 10^{32}$, gives $q < 90$, contradicting $q \geqslant 96$. We conclude that

$$q < 96. \tag{183}$$

The remaining indices are exactly

$$q = 3, 6, \ldots, 93, \qquad \sigma \in \{1, -1\}. \tag{184}$$

Because $\lambda > 2$, each strict factor-two window contains at most one integral power. Exact recurrence arithmetic constructs every candidate in equation (184), computes its coefficient content, and tests the norm identity. The complete terminal list is empty. This contradiction proves the theorem. □

Together with the even branch and the direct $q = 1$ check in equation (142), the odd exclusion proves theorem 1.1.

The order of the argument is essential. The fixed Matveev estimate first controls $\min\{h, q\gamma\}$; only in the large-q alternative does this become a bound for h. The moving estimate then bounds q, after which the same fixed and moving rational-interval certificates reduce the problem to the finite domain equation (184). No finite search is extrapolated beyond its proved range.

The all-exponent nearest-gap classification is now complete. We next reinterpret it dynamically, where it becomes the arithmetic input to a complete Pell-orbit theorem.

7. Arithmetic dynamics and complete Pell orbits

The nearest-gap equation is not an isolated algebraic coincidence. It is the exact image of an integral dynamical system under a quadratic-unit semiconjugacy. This section first proves that structural statement for a general seed, then specialises it to the two Fibonacci cores and obtains the complete list in theorem 1.2.

For an integral seed $z = (u, v)$, put

$$\begin{aligned} \mathsf{S}(u,v) &= (v, u+2v), \\ \mathcal{H}(u,v) &= u^2+v^2, \\ \mathcal{B}(u,v) &= v^2-u^2+2uv, \\ \mathcal{Q}(u,v) &= v^2-2uv-u^2, \end{aligned} \tag{185}$$

and define

$$A_z = (v-u) + u\sqrt{2}. \tag{186}$$

Proposition 7.1 (Seed/unit semiconjugacy)**.** *The map $z \mapsto A_z$ is a bijection*

$$\{(u,v) \in \mathbb{Z}^2 : 0 < u \leqslant v\} \longrightarrow \{a+b\sqrt{2} \in \mathbb{Z}[\sqrt{2}] : a \geqslant 0,\ b > 0\}, \tag{187}$$

with inverse $a + b\sqrt{2} \mapsto (b, a+b)$. For every integral seed and every $t \geqslant 0$,

$$N(A_z) = \mathcal{Q}(z), \qquad A_{\mathsf{S}^t z} = \lambda^t A_z. \tag{188}$$

Figure 4 displays the square that equation (188) commutes.

Proof. The inverse in equation (187) is immediate. Direct calculation gives

$$N(A_z) = (v-u)^2 - 2u^2 = \mathcal{Q}(z) \tag{189}$$

and

$$A_{\mathsf{S}z} = (u+v) + v\sqrt{2} = \lambda\big((v-u) + u\sqrt{2}\big). \tag{190}$$

Iteration proves equation (188). □

For a positive integral seed, $\mathcal{Q}(z) = 0$ would force $(v-u)^2 = 2u^2$, which is impossible. Thus the canonical case $\mathcal{Q}(z)^2 = 1$ and the noncanonical case $\mathcal{Q}(z)^2 > 1$ exhaust all positive seeds.

In coefficient columns this intertwining is an integral conjugacy. Indeed,

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \tag{191}$$

are respectively the seed step, multiplication by λ, and the map $(u,v)^{\mathsf{T}} \mapsto (v-u, u)^{\mathsf{T}}$. Direct multiplication gives

$$RM = UR. \tag{192}$$

Since $R \in \mathrm{GL}_2(\mathbb{Z})$, no lattice information is lost: $M^2 \in \mathrm{SL}_2(\mathbb{Z})$ is conjugate to the hyperbolic norm-one unit $\iota(\lambda^2)$.

$$\begin{tikzcd}
\begin{array}{c} z=(u,v) \\ \text{integral seed} \end{array} \arrow[r, "\mathsf{S}^t"] \arrow[d, "A"'] & \begin{array}{c} \mathsf{S}^t z \\ \text{evolved seed} \end{array} \arrow[d, "A"] \\
\begin{array}{c} A_z \in \mathbb{Z}[\sqrt{2}] \\ \text{quadratic integer} \end{array} \arrow[r, "\times \lambda^t"'] & \begin{array}{c} \lambda^t A_z \\ \text{unit translate} \end{array}
\end{tikzcd}$$

FIGURE 4. The exact seed/unit commutative diagram. The lower dynamics is multiplication in $\mathbb{Z}[\sqrt{2}]$, not an asymptotic model of the upper dynamics. Norm gives the signed covariance $\mathcal{Q}(\mathsf{S}^t z) = (-1)^t \mathcal{Q}(z)$; equivalently, $\mathcal{Q}^2$ is invariant.

There is a second, symmetric-square description. Direct expansion gives

$$\mathcal{B}(z)^2 + \mathcal{Q}(z)^2 = 2\mathcal{H}(z)^2, \tag{193}$$

and

$$\begin{pmatrix} \mathcal{H}(\mathsf{S}^t z) \\ \mathcal{B}(\mathsf{S}^t z) \\ \mathcal{Q}(\mathsf{S}^t z) \end{pmatrix} = \begin{pmatrix} C_{2t} & P_{2t} & 0 \\ 2P_{2t} & C_{2t} & 0 \\ 0 & 0 & (-1)^t \end{pmatrix} \begin{pmatrix} \mathcal{H}(z) \\ \mathcal{B}(z) \\ \mathcal{Q}(z) \end{pmatrix}. \tag{194}$$

The integral matrix in equation (194) preserves $2X^2 - Y^2 - Z^2$. Thus the quadratic observables of the seed lie on an integral Lorentz cone, and S acts through an arithmetic subgroup of its orthogonal group. This formulation is useful rather than decorative: the first two coordinates in equation (194) are exactly the proportionality mechanism behind the orbit/gap equivalence below.

Theorem 7.2 (Anchored orbit/gap equivalence)**.** *Let $z = (u, v)$ have positive integral coordinates and $\mathcal{Q}(z)^2 > 1$. Let d be positive and $t \geqslant 0$, and suppose either $u \leqslant v$ or $t \geqslant 1$. Then*

$$\mathcal{H}(\mathsf{S}^t z) = P_{d+2t} \tag{195}$$

if and only if there is an integer $g > 0$ such that

$$\lambda^{d-1} - A_z^2 = g\lambda^{-(2t+1)}. \tag{196}$$

At a hit, if

$$D = \mathcal{H}(z) - P_d, \qquad E = C_d - \mathcal{B}(z), \tag{197}$$

then

$$D = gP_{2t}, \qquad E = gC_{2t}, \qquad g = \gcd(D, E). \tag{198}$$

Moreover, equation (196) automatically satisfies

$$A_z^2 < \lambda^{d-1} < 2A_z^2, \tag{199}$$

and its coefficient content and norm are respectively g and $-g^2$.

Proof. Put $w = \mathsf{S}^t z$ and $\ell = d + 2t$. The hypotheses ensure that w is positive and ordered. At a hit, equation (193) and $C_\ell^2 - 2P_\ell^2 = (-1)^d$ give

$$\mathcal{B}(w)^2 = C_\ell^2 - \left(\mathcal{Q}(z)^2 + (-1)^d\right) < C_\ell^2. \tag{200}$$

Because w is positive and ordered, $0 < \mathcal{B}(w) < C_\ell$. For $t \geqslant 1$, comparison of the first coordinates in equation (194) with the Pell addition formula gives

$$C_{2t} D = P_{2t} E. \tag{201}$$

The second-coordinate difference is $\mathcal{B}(w) - C_\ell = -E/C_{2t}$, so $D, E > 0$. Since $\gcd(C_{2t}, P_{2t}) = 1$, this proves equation (198). At $t = 0$, the hit gives $D = 0$ and equation (200) gives $E > 0$; the same formula holds with $P_0 = 0$ and $C_0 = 1$.

The parity-free identity

$$\lambda A_z^2 = \mathcal{B}(z) + \mathcal{H}(z)\sqrt{2} \tag{202}$$

turns equation (197) into

$$\lambda^d - \lambda A_z^2 = E - D\sqrt{2}. \tag{203}$$

At a hit, equation (198) makes the right-hand side $g\lambda^{-2t}$, and division by λ proves equation (196). Conversely, equations (196) and (203) recover equation (198) coefficientwise; the first row of equation (194) then recovers the hit.

It remains to prove the window. Multiplying equation (196) by λ^{2t} and using equation (188) gives

$$\lambda^{\ell-1} - A_w^2 = g\lambda^{-1}. \tag{204}$$

At the hit, $g = C_\ell - \mathcal{B}(w)$. Combining this with equation (202) at w gives

$$\lambda A_w^2 - g = 2\mathcal{B}(w) + (-1)^{\ell+1}\lambda^{-\ell} > 0. \tag{205}$$

For even ℓ, positivity follows from $2\mathcal{B}(w) \geqslant 2 > \lambda^{-\ell}$; for odd ℓ it is immediate. Hence $0 < g\lambda^{-(2t+1)} < A_z^2$, which is equation (199). Finally, $\lambda^{-(2t+1)}$ is a coefficient-primitive norm-−1 unit, proving the content and norm assertions. ☐

Remark 7.3 (Canonical seeds)**.** If $0 < u \leqslant v$ and $\mathcal{Q}(z)^2 = 1$, then proposition 7.1 and the unit group of $\mathbb{Z}[\sqrt{2}]$ give a unique $r \geqslant 1$ with

$$z = (P_r, P_{r+1}), \qquad A_z = \lambda^r. \tag{206}$$

Consequently

$$\mathcal{H}(\mathsf{S}^t z) = P_{d+2t} \quad \Longleftrightarrow \quad d = 2r + 1, \tag{207}$$

and this equality persists for every $t \geqslant 0$. The corresponding gap is zero, so these seeds are correctly separated from theorem 7.2.

We now specialise to

$$z_{q,\sigma} = (F_{q+1}, F_{q+1} + \sigma F_q). \tag{208}$$

For both signs,

$$A_{z_{q,\sigma}} = D_{q,\sigma}, \qquad \mathcal{Q}(z_{q,\sigma}) = F_q^2 - 2F_{q+1}^2. \tag{209}$$

Theorem 7.4 (Complete all-anchor Fibonacci-core orbits)**.** *For every $q \geqslant 1$, $\sigma \in \{1, -1\}$, positive d, and $t \geqslant 0$, the hit*

$$\mathcal{H}(\mathsf{S}^t z_{q,\sigma}) = P_{d+2t} \tag{210}$$

occurs exactly in the four rows of equation (11).

Proof. First let $q \geqslant 2$. Then

$$\mathcal{Q}(z_{q,\sigma})^2 - 1 = F_{q-1}F_{q+1}F_{q+2}F_{q+4} > 0. \tag{211}$$

The strict window in theorem 7.2 first forces $d \geqslant 2$: if $d = 1$, it would give $A^2_{z_{q,\sigma}} < 1$, whereas $A_{z_{q,\sigma}} > 1$ for $q \geqslant 2$. Hence an odd anchor here has $d \geqslant 3$, so $d - 1$ is a positive even exponent.

For the plus sign the seed is ordered, so theorem 7.2 applies at every time. If d is odd, the gap exponent $d - 1$ is even. The only plus target in theorem 1.1 has normalised direction λ, whereas the orbit bridge requires $\lambda^{-(2t+1)}$. Hence there is no plus hit. If d is even, the gap exponent is odd, which is impossible by theorem 1.1. Thus the plus core has no hit for $q \geqslant 2$.

For the minus sign and $t \geqslant 1$, the evolved seed is ordered, so the same bridge applies. At an odd anchor, the only minus target has normalised direction λ^{-1}; it would force $t = 0$, a contradiction. At an even anchor, the exponent is odd and no target exists. Thus the minus core has no positive-time hit.

It remains to classify direct minus hits. The identity

$$\mathcal{H}(z_{q,-1}) = F^2_{q+1} + F^2_{q-1} = 3F^2_q + 2(-1)^q \tag{212}$$

and the complete Pell-value classification used in proposition 3.2 give exactly

$$(q, d) = (2, 3), (4, 5) \tag{213}$$

for $q \geqslant 2$. These are the last two rows of equation (11).

Finally, $z_{1,1} = (1, 2) = (P_1, P_2)$ is canonical. By remark 7.3, it gives $d = 3$ and every $t \geqslant 0$. For the minus sign, $z_{1,-1} = (1, 0)$ gives the direct hit $\mathcal{H}(1, 0) = P_1$. If $t \geqslant 1$, then

$$\mathsf{S}^t(1, 0) = (P_{t-1}, P_t), \qquad \mathcal{H}(\mathsf{S}^t(1, 0)) = P_{2t-1} < P_{d+2t}, \tag{214}$$

so there are no further hits. This proves the exhaustive list. □

Remark 7.5 (Exact scope)**.** The theorem classifies both signs, every positive anchor, and every nonnegative time for the displayed Fibonacci cores. It does not classify negative time or all noncanonical positive seeds with $\mathcal{Q}(z)^2 > 1$. Most importantly, no result here shows that a hypothetical Markoff collision must produce one of these cores. The semiconjugacy closes the downstream arithmetic problem; it does not supply the missing upstream Markoff reduction.

The next section changes perspective. Instead of classifying the global equation, it asks how much of its divisibility can recur locally, and thereby identifies the precise limits of p-adic information.

8. Local clocks, Haar recurrence, and their limits

The global classification uses the full norm–content equation. A tempting alternative is to control only the common prime-power content forced by the Fibonacci and Pell recurrences. We now show exactly how far that local idea can go. Compatible p-adic clocks recur on positive-density subfamilies of genuine nearest windows, but a rank-parity obstruction can also prevent a split prime from occurring at all.

For the plus sign put

$$D_q = F_q + F_{q+1}\sqrt{2}, \qquad m(q) = \lceil \log_\lambda D_q \rceil . \tag{215}$$

Whenever $3 \nmid q$ and

$$D_q < \lambda^{m(q)} < \sqrt{2} D_q, \tag{216}$$

the even power $\lambda^{2m(q)}$ lies in the strict nearest window for D_q^2. Put $s(m) = (-1)^m$ and define

$$G(q, m) = \gcd\left(R_q - s(m),\, 2F_{q+1}P_m - F_qC_m\right). \tag{217}$$

This cross-gcd is defined on every row in equation (216); the norm–content equation is not assumed.

Lemma 8.1 (Two exact phase factorizations)**.** *For $3 \nmid q$,*

$$\begin{aligned} m \equiv q \pmod 2 \quad &\Longrightarrow \quad G(q, m) = \gcd(F_{q+1}, C_m)\gcd(F_{q+2}, C_{m+1}) \\ m \not\equiv q \pmod 2 \quad &\Longrightarrow \quad G(q, m) = \gcd(F_{q-1}, C_{m-1})\gcd(F_{q+4}, C_{m+2}). \end{aligned} \tag{218}$$

Proof. Put $\varepsilon = (-1)^q$ and $E = 2F_{q+1}P_m - F_qC_m$. Cassini's identity gives

$$\begin{aligned} R_q - \varepsilon &= F_{q+1}F_{q+2}, \\ R_q + \varepsilon &= F_{q-1}F_{q+4}. \end{aligned} \tag{219}$$

Direct reduction gives

$$\begin{aligned} E &\equiv -F_qC_m &&\pmod{F_{q+1}}, \\ E &\equiv -F_qC_{m+1} &&\pmod{F_{q+2}}, \\ E &\equiv F_qC_{m-1} &&\pmod{F_{q-1}}, \\ 3E &\equiv -F_qC_{m+2} &&\pmod{F_{q+4}}. \end{aligned} \tag{220}$$

The equal-phase Fibonacci factors are coprime. In the unequal phase,

$$\gcd(F_{q-1}, F_{q+4}) \in \{1, 5\},$$

but the exceptional prime 5 divides no companion number C_j: the recurrence modulo 5 has period 12 and nonzero residues $1, 1, 3, 2, 2, 1, 4, 4, 2, 3, 3, 4$. The only possible failure of the displayed multipliers to be units is at 3. Here

$$3 \mid F_r \Longleftrightarrow 4 \mid r, \qquad 3 \mid C_j \Longleftrightarrow j \equiv 2 \pmod 4,$$

and every aligned Fibonacci–companion index pair has opposite parity. Thus 3 cannot occur in an aligned gcd. The product identity follows prime by prime and valuation by valuation. □

The factorisation replaces one two-coordinate gcd by two aligned recurrence clocks. We now determine exactly which primes can synchronise in one of these clocks to arbitrary depth.

Let p be an odd prime with $(2/p) = 1$ that divides some C_j. Define

$$\begin{aligned} z_F(p) &= \min\{r > 0 : p \mid F_r\}, \quad e_F(p) = v_p(F_{z_F(p)}), \\ z_C(p) &= \min\{r > 0 : p \mid C_r\}, \quad e_C(p) = v_p(C_{z_C(p)}), \end{aligned} \tag{221}$$

and, for $a \geqslant 1$,

$$B_p(a) = z_F(p)p^{\max(0, a - e_F(p))}, \qquad J_p(a) = z_C(p)p^{\max(0, a - e_C(p))}. \tag{222}$$

The exact rank laws are

$$p^a \mid F_r \Longleftrightarrow B_p(a) \mid r, \qquad p^a \mid C_j \Longleftrightarrow j \equiv J_p(a) \pmod{2J_p(a)}. \tag{223}$$

The first is the standard Fibonacci rank and valuation law [14, 21]. For the second, choose a Hensel lift $\sqrt{2} \in \mathbb{Z}_p$, put $u = 1 + \sqrt{2}$ and $b = -u^2$, and use

$$2u^jC_j = (-1)^j(b^j + 1). \tag{224}$$

Companion zeros are the odd multiples of $z_C(p)$, and odd-prime lifting gives their complete valuations. In local-field language, $\mathbb{Q}(\sqrt{2}) \otimes \mathbb{Q}_p \simeq \mathbb{Q}_p \times \mathbb{Q}_p$ at such a split prime, and equation (224) converts the clock into an order and valuation statement for a unit of $\mathbb{Z}_p$; this is the special case of the standard framework described by Serre [22].

Proposition 8.2 (Local-to-Archimedean transfer)**.** *Let $\mathcal{P}$ be a finite nonempty set of primes satisfying the preceding hypotheses, prescribe $a_p \geqslant 1$, and partition $\mathcal{P} = \mathcal{P}_1 \sqcup \mathcal{P}_2$. Choose $\delta \in \{0, 1\}$ and the offsets*

δ	(r_1, t_1)	(r_2, t_2)
0	$(1, 0)$	$(2, 1)$
1	$(-1, -1)$	$(4, 2)$

(225)

Suppose that integers q_0, m_0 satisfy

$$3 \nmid q_0, \qquad q_0 - m_0 \equiv \delta \pmod 2, \tag{226}$$

and, for $i \in \{1, 2\}$ and $p \in \mathcal{P}_i$,

$$q_0 + r_i \equiv 0 \pmod{B_p(a_p)}, \qquad m_0 + t_i \equiv J_p(a_p) \pmod{2J_p(a_p)}. \tag{227}$$

Put

$$Q = \operatorname{lcm}\left(6, \{B_p(a_p) : p \in \mathcal{P}\}\right), \qquad M = \operatorname{lcm}\left(\{2J_p(a_p) : p \in \mathcal{P}\}\right). \tag{228}$$

Then infinitely many $q = q_0 + Q\ell$ satisfy equation (216), $m(q) \equiv m_0 \pmod M$, and

$$\prod_{p \in \mathcal{P}} p^{a_p} \mid G(q, m(q)), \tag{229}$$

with every prime power carried by its prescribed factor in equation (218). Among the progression $q = q_0 + Q\ell$, this selected family has relative natural density

$$\frac{\vartheta}{M}, \qquad \vartheta = \log_\lambda \sqrt{2}. \tag{230}$$

Its counting function up to $q \leqslant X$ is therefore $\vartheta X/(MQ) + o(X)$.

Proof. The congruences and equation (223) place each prescribed prime power in the required Fibonacci–companion pair of equation (218). They remain valid on the progression $q = q_0 + Q\ell$.

It remains to meet the Archimedean window with the required value of m. Put

$$\gamma = \frac{\log \varphi}{\log \lambda}, \qquad \beta = \log_\lambda \left(\frac{1 + \sqrt{2}\varphi}{\sqrt{5}}\right). \tag{231}$$

Binet's formula gives

$$\log_\lambda D_q = q\gamma + \beta + O(\varphi^{-2q}). \tag{232}$$

The irrationality of γ was proved in theorem 2.3; hence $Q\gamma/M$ is irrational. Weyl equidistribution on the circle $\mathbb{R}/M\mathbb{Z}$ shows that the limiting phase lies in the cyclic interval $(m_0 - \vartheta, m_0)$ with normalised Haar measure ϑ/M. Its boundary has Haar measure zero, and the Binet error tends to zero. Membership in this interval is exactly $m(q) \equiv m_0 \pmod M$ together with equation (216). This proves the divisibility and both density assertions. ∎

Theorem 8.3 (Exact clock criterion and simultaneous products)**.** *Let p be an odd prime with $(2/p) = 1$ that divides some C_j. Then the following are equivalent:*

(i) *$z_F(p)$ and $z_C(p)$ are not both even;*
(ii) *for every $a \geqslant 1$, infinitely many nearest-window rows have $p^a \mid G(q, m(q))$;*
(iii) *at least one such row has $p \mid G(q, m(q))$.*

Every prime $p \equiv 7 \pmod 8$ satisfies these conditions. Consequently, for every finite set $\mathcal{P}$ of such primes and arbitrary exponents $a_p \geqslant 1$, infinitely many rows satisfy the one-bin divisibility

$$\prod_{p\in\mathcal{P}} p^{a_p} \mid \gcd(F_{q+1}, C_{m(q)}) \mid G(q, m(q)). \tag{233}$$

The criterion also reaches primes outside that residue class: $z_F(17) = 9$ is odd, so 17 satisfies condition (i) although $17 \equiv 1 \pmod 8$. Assigning 7 to $\gcd(F_{q+1}, C_{m(q)})$ and 17 to $\gcd(F_{q+2}, C_{m(q)+1})$, that is to the two different factors of lemma 8.1, gives, for every $a, b \geqslant 1$, infinitely many rows with

$$7^a 17^b \mid G(q, m(q)). \tag{234}$$

The rank-parity condition is sharp: $z_F(241) = 120$ and $z_C(241) = 20$, so

$$241 \nmid G(q, m) \tag{235}$$

for every plus nearest-window row.

Proof. For a single prime, use the first factor in the appropriate phase and write $q = B_p(a)u - 1$, with $m \equiv J_p(a) \pmod{2J_p(a)}$. The phase condition becomes

$$B_p(a)u \equiv J_p(a) + 1 \pmod 2. \tag{236}$$

If $B_p(a)$ is even, this is soluble exactly when $J_p(a)$ is odd; if $B_p(a)$ is odd, the parity of u is free. Multiplication by powers of the odd prime does not change the rank parities, so the congruences are consistent exactly when $z_F(p)$ and $z_C(p)$ are not both even. Choose the residue of u modulo 3 so that $3 \nmid B_p(a)u - 1$. If $3 \mid B_p(a)$ this is automatic; otherwise exactly one residue is forbidden. This choice is compatible with the required parity by the Chinese remainder theorem modulo 6. Proposition 8.2 then gives infinitely many rows at every depth. Conversely, the Fibonacci and companion indices paired in equation (218) have opposite parity. If both ranks are even, no pair can contain p. This proves the equivalence.

Now let $p \equiv 7 \pmod 8$ and choose r with $r^2 \equiv 2 \pmod p$. For $\pi = 1 + r$, the order of π has 2-part at most 2. If e is its odd part, then $\pi^e = \pm 1$; since $1 - r = -\pi^{-1}$, this gives $C_e \equiv 0 \pmod p$. Thus $z_C(p)$ is odd, and the criterion applies. The assertion for $\mathcal{P} = \varnothing$ follows from theorem 2.3, so suppose henceforth that $\mathcal{P} \neq \varnothing$. For finitely many such primes, the lifted companion ranks remain odd. Put

$$B = \mathrm{lcm}_{p\in\mathcal{P}}\, B_p(a_p), \qquad J = \mathrm{lcm}_{p\in\mathcal{P}}\, J_p(a_p).$$

Then $m \equiv J \pmod{2J}$ satisfies every odd companion class, and $q = Bu - 1$ satisfies every first-bin Fibonacci class. Choose the parity of u to make $q \equiv m \pmod 2$, and choose its residue modulo 3 as above. The choices are compatible modulo 6, so proposition 8.2 proves equation (233). There are infinitely many primes $7 \pmod 8$: if $p_1, \ldots, p_k$ were all of them, an even multiple A of their product would make $2A^2 - 1 \equiv 7 \pmod 8$; every prime divisor r has $(2/r) = 1$, hence $r \equiv 1$ or $7 \pmod 8$, and at least one must be $7 \pmod 8$. It is new because $2A^2 - 1 \equiv -1 \pmod{p_i}$ for every i.

For equation (234), the exact lifted ranks are

$$\begin{aligned} B_7(a) &= 8 \cdot 7^{a-1}, \quad J_7(a) = 3 \cdot 7^{a-1}, \\ B_{17}(b) &= 9 \cdot 17^{b-1}, \quad J_{17}(b) = 4 \cdot 17^{b-1}. \end{aligned} \tag{237}$$

Assign 7 and 17 to the two equal-phase bins. Their q moduli are coprime and their m moduli have gcd 2 with odd prescribed residues, so the systems are compatible. More explicitly, $q \equiv -1 \pmod{B_7(a)}$ makes q odd, both companion congruences make m odd, and $q \equiv -2 \pmod{B_{17}(b)}$ gives $q \equiv 1 \pmod 3$ because $9 \mid B_{17}(b)$. Thus the phase and $3 \nmid q$ hypotheses of proposition 8.2 hold. Finally, the exact recurrence certificates at 241 give the two even ranks displayed in the theorem; the already proved criterion yields equation (235). ☐

Remark 8.4 (What the clocks establish)**.** The rows above satisfy genuine strict nearest windows, and the density is a Haar-measure statement on a phase circle. They are not asserted to satisfy equation (24). Thus local abundance produces neither a target nor a Markoff counterexample. Its exact conclusion is methodological: uniform boundedness, squarefreeness, finitely many compatible rank clocks, and the assertion that every split prime occurs are all false as global substitutes for the norm–content equation.

This local–global contrast is the organising principle of the final discussion, which situates the exact theorems relative to the parent Markoff problem.

9. Why the Fibonacci and Pell arms

A reader who has followed the proof may reasonably ask why this particular pair of sequences carries the whole argument. The Markoff tree has infinitely many branches; Fibonacci and Pell are two of them. This section answers the question without claiming a global classification of branch dynamics. Fibonacci and Pell are the two constant-directive branches, their integral half-step actions are explicit, and the constant γ of equation (33) is the exchange rate between their costs. Any analogous action on another branch must pass a stringent trace obstruction.

Nothing in this section is used in the proofs of theorems 1.1, 1.2 and 8.3. It is included because it explains their shape, and because it delimits the generalisations that can and cannot be attempted.

We are careful throughout to separate what is proved from what is only verified on a finite family. Finite checks are explicitly identified as such and are never used as hypotheses.

9.1. Background from the Christoffel–Markoff correspondence.

We use the classical dictionary between Christoffel words and Markoff numbers in the form given by Reutenauer [19], whose origin is Frobenius's observation of 1913 [9] and Cohn's matrix formulation. The one normalisation we need beyond the correspondence itself is $m = \frac{1}{3}\operatorname{tr} M_w = (M_w)_{12}$. In the transposed convention this is [18, Lemme 3.2], which states that $3N_{21} = \operatorname{tr} N$ for the corresponding conjugate; the supplement also verifies the identity directly, in the convention used here, on every Christoffel word of depth at most eleven.

Throughout this section, a *normalized Markoff occurrence* means the representative encoded by the proper lower Christoffel word $a\operatorname{Pal}(v)b$ at a reduced Farey index $0 < P < Q$, where Pal is iterated right palindromic closure, after removing the reversal and letter-exchange symmetries. This convention fixes the orientation whenever an occurrence-induced two-square writing is decoded back to a word.

Let

$$\mathsf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathsf{B} = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \tag{238}$$

and for a word w over $\{a, b\}$ write M_w for the corresponding product, $a \mapsto \mathsf{A}$ and $b \mapsto \mathsf{B}$. If w is a Christoffel word, then $M_w \in \mathrm{SL}_2(\mathbb{Z})$ has positive entries and

$$M_w = \begin{pmatrix} A & m \\ C & D \end{pmatrix}, \qquad A + D = 3m, \qquad 1 \leqslant D \leqslant m, \tag{239}$$

where m is the Markoff number attached to w. Two consequences are used below. First, $AD - mC = 1$ and $A \equiv -D \pmod{m}$ give $D^2 \equiv -1 \pmod{m}$, so the residue

$$u = A - 2m = m - D \tag{240}$$

lies in $[0, m)$ and satisfies $u^2 + 1 \equiv 0 \pmod{m}$; it is the characteristic root attached to the occurrence w. Second, the Markoff word attached to this occurrence, namely the Christoffel word of slope $u/(m - u)$, has length exactly m, with

$$m_b = u \text{ letters } b, \qquad m_a = m - u \text{ letters } a. \tag{241}$$

We write $\varphi = (1 + \sqrt{5})/2$ throughout, as elsewhere in this paper.

9.2. **The writing and the letter counts determine each other.** The seed data of this paper is a primitive two-square writing $m = p^2 + q^2$, whereas the word side supplies the letter counts equation (241). The passage between the two is the Gauss correspondence between primitive writings and square roots of -1, and it takes the concrete form of a congruence with an explicit quotient. We record it because it turns the first row of table 1 into an identity rather than an analogy, and because the orientation ambiguity it carries is exactly the ambiguity $u \leftrightarrow m - u$ of the characteristic root.

Proposition 9.1 (Gauss congruence and its orientation)**.** *Let p and q be coprime positive integers, put $m = p^2 + q^2$, and let $u \in [0, m)$ be the residue determined by $uq \equiv p \pmod{m}$. Then*

(i) $u^2 + 1 \equiv 0 \pmod{m}$*;*

(ii) $q - p + u(p + q) \equiv 0 \pmod{m}$*, and the quotient*

$$k = \frac{u(p + q) + q - p}{m} \tag{242}$$

is the unique integer of the open interval $(0, p+q)$ satisfying $kq \equiv 1 \pmod{p+q}$;

(iii) *exchanging p and q replaces u by $m - u$ and k by $p + q - k$; the companion congruence is $p - q + (m - u)(p + q) \equiv 0 \pmod{m}$, and its quotient $p + q - k$ is the inverse of p modulo $p + q$.*

Proof. The residue is well defined. A common divisor of q and $m = p^2 + q^2$ divides p^2, hence divides $\gcd(p, q)^2 = 1$. So q is invertible modulo m, and $u = pq^{-1} \bmod m$ exists and is unique in $[0, m)$.

Part (i). Reducing $m = p^2 + q^2$ modulo m gives $p^2 \equiv -q^2$. Multiplying by the square of q^{-1},

$$u^2 = p^2\left(q^{-1}\right)^2 \equiv -q^2\left(q^{-1}\right)^2 = -1 \pmod{m}. \tag{243}$$

Part (ii), the congruence. Expanding and using $p^2 \equiv -q^2$ once,

$$u(p + q) \equiv pq^{-1}(p + q) = p + p^2 q^{-1} \equiv p - q^2 q^{-1} = p - q \pmod{m}, \tag{244}$$

which is the assertion.

Part (ii), the quotient lies in $(0, p+q)$. First $u \neq 0$: otherwise $p \equiv 0 \pmod{m}$, which is impossible because $1 \leqslant p < m$. Hence $1 \leqslant u \leqslant m-1$, and the numerator of equation (242) is bounded on both sides by

$$2q = (p+q) + q - p \leqslant u(p+q) + q - p \leqslant (m-1)(p+q) + q - p = m(p+q) - 2p. \quad (245)$$

The left end is positive and the right end is smaller than $m(p+q)$, so dividing by m gives $0 < k < p+q$; and k is an integer because the numerator is divisible by m by the previous step.

Part (ii), the quotient is an inverse. By the definition of u there is an integer a with $uq = p + am$. Multiplying $km = u(p+q) + q - p$ by q and substituting,

$$kmq = uq(p+q) + q^2 - pq = (p+am)(p+q) + q^2 - pq = p^2 + q^2 + am(p+q), \quad (246)$$

the term pq cancelling. The right-hand side is $m + am(p+q)$, so dividing through by m,

$$kq = 1 + a(p+q). \quad (247)$$

Thus $kq \equiv 1 \pmod{p+q}$. Since $\gcd(q, p+q) = \gcd(q,p) = 1$, that inverse is unique modulo $p+q$, and by the previous step k is its representative in $(0, p+q)$.

Part (iii). Exchanging p and q replaces u by the residue u' determined by $u'p \equiv q$, that is $u' = qp^{-1} \bmod m$. Then $uu' \equiv 1$, and multiplying $u^2 \equiv -1$ by u' gives $u \equiv u^2 u' \equiv -u'$; since both lie in $[0, m)$ and $u \neq 0$, this means $u' = m - u$. Parts (i) and (ii) applied to the pair (q,p) give the companion congruence together with the statement that its quotient $k' = \big(u'(p+q) + p - q\big)/m$ is the inverse of p modulo $p+q$. Finally, adding the two numerators and using $u + u' = m$,

$$(k+k')m = (u+u')(p+q) = m(p+q), \quad (248)$$

so $k' = p + q - k$. □

Proposition 9.2 (The writing attached to a root of -1). *Let $m \geqslant 5$ and let u be an integer with $u^2 + 1 \equiv 0 \pmod{m}$. Then there is exactly one pair (p,q) of positive integers with $m = p^2 + q^2$ and $p \equiv uq \pmod{m}$, and that pair is coprime.*

Proof. A small solution of the congruence. Put $n = \lfloor \sqrt{m} \rfloor$ and consider the $(n+1)^2$ pairs (s,t) of integers with $0 \leqslant s,t \leqslant n$. Since $(n+1)^2 > m$, two distinct such pairs (s_1, t_1) and (s_2, t_2) satisfy $s_1 - ut_1 \equiv s_2 - ut_2 \pmod{m}$. Put $x = s_1 - s_2$ and $y = t_1 - t_2$, so that

$$x \equiv uy \pmod{m}, \qquad |x| \leqslant n, \qquad |y| \leqslant n, \quad (249)$$

with $(x,y) \neq (0,0)$. Neither coordinate vanishes. If $y = 0$ then $m \mid x$ while $0 < |x| \leqslant n < m$; and if $x = 0$ then $m \mid uy$, hence $m \mid y$ because $\gcd(u,m) = 1$, while $0 < |y| \leqslant n < m$.

The pair represents m. Squaring equation (249) and using $u^2 \equiv -1$,

$$x^2 + y^2 \equiv u^2 y^2 + y^2 = (u^2+1)y^2 \equiv 0 \pmod{m}, \quad (250)$$

while $0 < x^2 + y^2 \leqslant 2n^2 \leqslant 2m$. So $x^2 + y^2$ is either m or $2m$. Suppose it were $2m$. Then $|x| = |y| = n$ and $m = n^2$, so $x = \delta n$ and $y = \delta' n$ for two signs $\delta, \delta' \in \{1, -1\}$, and $x - uy = n(\delta - u\delta')$. Since $m = n^2$ divides $x - uy$, we get $n \mid \delta - u\delta'$, hence $u \equiv \delta\delta' \pmod{n}$ and therefore $u^2 \equiv 1 \pmod{n}$. But $n \mid m$ and $m \mid u^2 + 1$ give $u^2 \equiv -1 \pmod{n}$, so $n \mid 2$ and $m = n^2 \leqslant 4$, contradicting $m \geqslant 5$. Hence $x^2 + y^2 = m$.

The pair is coprime. Let $g = \gcd(x, y)$ and write $x = gx_1$, $y = gy_1$. Then $g^2 \mid x^2+y^2 = m$; put $m_1 = m/g^2 = x_1^2 + y_1^2$. The divisibility $m \mid x - uy$ reads $g^2m_1 \mid g(x_1 - uy_1)$, that is

$$x_1 \equiv uy_1 \pmod{gm_1}. \tag{251}$$

Squaring it and using $u^2 \equiv -1$ modulo g^2m_1, hence a fortiori modulo gm_1,

$$m_1 = x_1^2 + y_1^2 \equiv (u^2 + 1)y_1^2 \equiv 0 \pmod{gm_1}. \tag{252}$$

A positive integer divisible by gm_1 and equal to m_1 forces $g = 1$.

Normalisation. Changing (x, y) into $(-x, -y)$ leaves equation (249) unchanged, so we may assume $y > 0$. If $x > 0$, take $(p, q) = (x, y)$. If $x < 0$, put $p = y$ and $q = -x$, both positive; multiplying $x \equiv uy$ by u and using $u^2 \equiv -1$ gives $ux \equiv -y$, that is $y \equiv u(-x)$, which is $p \equiv uq$. In both cases $m = p^2 + q^2$ with $\gcd(p, q) = 1$ and $p \equiv uq \pmod{m}$.

Uniqueness. Let (p, q) and (P, Q) be pairs of positive integers with $m = p^2 + q^2 = P^2 + Q^2$, $p \equiv uq$ and $P \equiv uQ$ modulo m. Then

$$pQ - qP \equiv uqQ - quQ = 0 \pmod{m}. \tag{253}$$

On the other hand $pQ - qP$ is the inner product of (p, q) with $(Q, -P)$, so by the Cauchy–Schwarz inequality

$$|pQ - qP| \leqslant \sqrt{p^2 + q^2}\,\sqrt{P^2 + Q^2} = m, \tag{254}$$

with equality only if (p, q) and $(Q, -P)$ are proportional. That is impossible: $p = \mu Q$ would need $\mu > 0$ and $q = -\mu P$ would need $\mu < 0$. Hence $|pQ - qP| < m$, and the congruence forces $pQ = qP$. Since $\gcd(p, q) = 1$, this gives $P = \kappa p$ and $Q = \kappa q$ for a positive integer κ, and then $m = P^2 + Q^2 = \kappa^2 m$ gives $\kappa = 1$. ☐

Corollary 9.3 (An occurrence's writing and letter counts determine each other)**.** *Let w be a Christoffel word with Markoff value $m \geqslant 5$, and let u be the characteristic root of this occurrence from equation (240). Then m has exactly one writing $m = p^2 + q^2$ in positive integers with $p \equiv uq \pmod{m}$, that writing is primitive, and the letter counts equation (241) of the Markoff word attached to w satisfy*

$$m_b(p + q) \equiv p - q, \qquad m_a(p + q) \equiv q - p \pmod{m}. \tag{255}$$

Exchanging p with q replaces u by $m - u$ and exchanges m_a with m_b.

Proof. The characteristic root satisfies $u^2 + 1 \equiv 0 \pmod{m}$ by equation (240), so proposition 9.2 supplies the writing, its primitivity and its uniqueness. Its defining congruence $p \equiv uq$ says that the residue attached to (p, q) in proposition 9.1 is u itself, so equation (244) gives $u(p + q) \equiv p - q$; by equation (241) that residue is the letter count m_b, which is the first congruence. The second follows from $m_a = m - m_b$, since then $m_a(p + q) \equiv -u(p + q) \equiv q - p$. The last assertion is part (iii) of proposition 9.1. ☐

For a fixed normalized Markoff occurrence, the two-square writing and the Markoff word are therefore two readings of one object, and each is computed from the other: the writing from the occurrence's characteristic root by proposition 9.2, the letter counts from the writing by equation (255). Within that occurrence, the orientation ambiguity is the exchange $u \leftrightarrow m - u$, which is the choice of an ordering of the pair (p, q). The next subsection computes both sides explicitly on the two extreme arms of the tree, where the letter counts turn out to be the coordinates of a power of the fundamental unit.

9.3. **Letter counts on the two arms are unit coordinates.** The leftmost branch of the Christoffel tree consists of the words $a^k b$ and produces the Fibonacci Markoff numbers; the rightmost consists of the words ab^k and produces the Pell Markoff numbers. On both, the letter counts are not merely reminiscent of the fundamental unit: they are its coordinates.

Proposition 9.4 (Letter counts on the two arms)**.** *For $k \geqslant 1$,*

$$(m_a, m_b) = (F_{2k}, F_{2k-1}) \quad \textit{when} \quad m = F_{2k+1}, \tag{256}$$

and

$$(m_a, m_b) = (C_{2k}, P_{2k}) \quad \textit{when} \quad m = P_{2k+1}. \tag{257}$$

Equivalently, (m_a, m_b) is the coordinate vector of φ^{2k} in the basis $(\varphi, 1)$ of $\mathbb{Z}[\varphi]$ in the first case, and of λ^{2k} in the basis $(1, \sqrt{2})$ of $\mathbb{Z}[\sqrt{2}]$ in the second.

Proof. Both cases are a direct computation with the Cohn generators equation (238), using equation (240) in the form $m = (M_w)_{12}$ and $u = (M_w)_{11} - 2m$. Two elementary matrix identities drive them. Writing

$$\mathsf{F} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathsf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \tag{258}$$

one checks at once that $\mathsf{F}^2 = \mathsf{A}$ and $\mathsf{P}^2 = \mathsf{B}$, and the standard induction gives

$$\mathsf{F}^r = \begin{pmatrix} F_{r+1} & F_r \\ F_r & F_{r-1} \end{pmatrix}, \qquad \mathsf{P}^r = \begin{pmatrix} P_{r+1} & P_r \\ P_r & P_{r-1} \end{pmatrix}. \tag{259}$$

Fibonacci arm. The word is $w = a^{k-1}b$, so $M_w = \mathsf{A}^{k-1}\mathsf{B} = \mathsf{F}^{2k-2}\mathsf{B}$. Multiplying equation (259) by B gives

$$M_w = \begin{pmatrix} 5F_{2k-1} + 2F_{2k-2} & 2F_{2k-1} + F_{2k-2} \\ 5F_{2k-2} + 2F_{2k-3} & 2F_{2k-2} + F_{2k-3} \end{pmatrix}. \tag{260}$$

Its $(1, 2)$ entry is $2F_{2k-1} + F_{2k-2} = F_{2k-1} + (F_{2k-1} + F_{2k-2}) = F_{2k-1} + F_{2k} = F_{2k+1}$, so $m = F_{2k+1}$. Subtracting twice this from the $(1, 1)$ entry,

$$u = (5F_{2k-1} + 2F_{2k-2}) - 2(2F_{2k-1} + F_{2k-2}) = F_{2k-1}. \tag{261}$$

Hence $m_b = F_{2k-1}$ and $m_a = F_{2k+1} - F_{2k-1} = F_{2k}$, which is equation (256). The coordinate statement is Binet's identity $\varphi^{2k} = F_{2k}\varphi + F_{2k-1}$, immediate by induction from $\varphi^2 = \varphi + 1$.

Pell arm. The word is $w = ab^k$, so $M_w = \mathsf{A}\mathsf{B}^k = \mathsf{A}\mathsf{P}^{2k}$, and equation (259) gives

$$M_w = \begin{pmatrix} 2P_{2k+1} + P_{2k} & 2P_{2k} + P_{2k-1} \\ P_{2k+1} + P_{2k} & P_{2k} + P_{2k-1} \end{pmatrix}. \tag{262}$$

Its $(1, 2)$ entry is $2P_{2k} + P_{2k-1} = P_{2k+1}$ by the Pell recursion, so $m = P_{2k+1}$. Subtracting twice this from the $(1, 1)$ entry,

$$u = (2P_{2k+1} + P_{2k}) - 2P_{2k+1} = P_{2k}. \tag{263}$$

Hence $m_b = P_{2k}$ and $m_a = P_{2k+1} - P_{2k} = C_{2k}$, the last equality being the recursion $P_{r+1} = C_r + P_r$ recorded after equation (16). This is equation (257).

Both computed roots do satisfy $u^2 + 1 \equiv 0$ modulo m, as equation (240) requires, and in both cases this is a determinant. Since $\det \mathsf{F} = \det \mathsf{P} = -1$, taking determinants in equation (259) at the even exponent $r = 2k$ gives

$$F_{2k+1}F_{2k-1} - F_{2k}^2 = 1, \qquad P_{2k+1}P_{2k-1} - P_{2k}^2 = 1. \tag{264}$$

The second congruence is immediate: $P_{2k}^2 \equiv -1 \pmod{P_{2k+1}}$. The first gives $F_{2k}^2 \equiv -1 \pmod{F_{2k+1}}$, and since $F_{2k-1} = F_{2k+1} - F_{2k} \equiv -F_{2k}$, also $F_{2k-1}^2 \equiv F_{2k}^2 \equiv -1 \pmod{F_{2k+1}}$. □

The same two arms also exhibit the square mechanism that first appeared in equation (64). If $U_0 = 0$, $U_1 = 1$, and $U_{r+1} = tU_r + U_{r-1}$, the addition formula for this recurrence gives

$$U_{2k+1} = U_k^2 + U_{k+1}^2. \tag{265}$$

At $t = 1$ this reads $F_{2k+1} = F_k^2 + F_{k+1}^2$, and at $t = 2$ it reads $P_{2k+1} = P_k^2 + P_{k+1}^2$. Every non-endpoint Markoff occurrence has an oriented primitive two-square writing by corollary 9.3. What is special here is stronger: on these two arms the summands are consecutive terms of the same second-order recurrence that produces the Markoff number. With $t = 1$ and $k = q + 1$, equation (265) is exactly the final equality in equation (64); the squares are therefore the hinge between the Markoff writing and the Fibonacci–Pell intersection, not an accidental simplification.

Remark 9.5 (A numerical coincidence explained)**.** The case $k = 7$ of equation (256) reads $m = F_{15} = 610$ and $(m_a, m_b) = (377, 233)$, and the corresponding unit identity is $\varphi^{14} = 377\varphi + 233$, which is equation (101). The two occurrences of the pair $(377, 233)$ in this paper, one as a threshold in the even-branch estimate and one as the letter counts of the Markoff word of 610, are therefore the same identity read twice. The first nontrivial Pell case is $\lambda^4 = 17 + 12\sqrt{2}$: the Markoff word of 29 has 17 letters a and 12 letters b.

For each non-endpoint normalized Markoff occurrence, the passage from its two-square writing to its letter counts is not an analogy to be tested on examples: corollary 9.3 makes it an identity. It can also be watched on the word side, where it takes a purely combinatorial form. Starting from the oriented two-square pair, subtractive Euclid recovers the half directive; iterated palindromic closure then constructs its Christoffel word, while reversal complement doubles the half directive. At $m = 29$ the pair $(2, 5)$ returns the half directive aab, whose reversal complement abb prefixes it into the directive word $abbaab$, and the Markoff word it builds has the 17 letters a and 12 letters b that equation (241) predicts from the characteristic root $u = 12$, a quantity that mentions neither the pair nor any palindrome.

9.4. **Fixed-entry branch monodromy has a quadratic Pisot eigenvalue.** Fix a Christoffel word w_0 with Markoff number m_0, and follow the branch of the tree that keeps m_0 fixed, so that the node words are $w_0^k v_0$ for a fixed v_0. Write m_k for the Markoff number of the k-th node and u_k for its characteristic root.

Proposition 9.6 (Branch recurrence and its Pisot eigenvalue)**.** *The vector* $(m_a, m_b)_k = (m_k - u_k, u_k)$ *satisfies*

$$(m_a, m_b)_{k+1} = 3m_0\,(m_a, m_b)_k - (m_a, m_b)_{k-1}. \tag{266}$$

Its characteristic polynomial is $x^2 - 3m_0x + 1$*, whose dominant root*

$$\beta(m_0) = \frac{3m_0 + \sqrt{9m_0^2 - 4}}{2} \tag{267}$$

is a quadratic Pisot unit, hence in particular a Perron number, lying in the order of discriminant $9m_0^2 - 4$*.*

Proof. Write $\mathsf{C}_k = M_{w_0^k v_0}$ for the Cohn matrix of the k-th node. Since the Cohn correspondence is multiplicative on concatenation, $\mathsf{C}_k = M_{w_0}^k M_{v_0}$.

Step 1: the matrix recurrence. The matrix M_{w_0} lies in $\mathrm{SL}_2(\mathbb{Z})$ and has trace $3m_0$ by equation (239). Its characteristic polynomial is therefore $x^2 - 3m_0 x + 1$, and Cayley–Hamilton gives $M_{w_0}^2 = 3m_0 M_{w_0} - I$. Multiplying on the right by $M_{w_0}^{k-1} M_{v_0}$ yields

$$\mathsf{C}_{k+1} = 3m_0 \mathsf{C}_k - \mathsf{C}_{k-1}, \tag{268}$$

an identity between matrices, hence between each of their four entries separately.

Step 2: the two entries we need. By equation (239), the Markoff number is the $(1,2)$ entry, $m_k = (\mathsf{C}_k)_{12}$, and the $(1,1)$ entry is $(\mathsf{C}_k)_{11}$. Both obey the scalar recurrence $x_{k+1} = 3m_0 x_k - x_{k-1}$, by Step 1.

Step 3: the characteristic root obeys it too. By equation (240), $u_k = (\mathsf{C}_k)_{11} - 2m_k$. A fixed integer linear combination of two solutions of a linear recurrence is again a solution, so u_k satisfies $x_{k+1} = 3m_0 x_k - x_{k-1}$. The constancy of the coefficient 2 is exactly the normalisation $1 \leqslant D \leqslant m$ recorded in equation (239), which holds at every node.

Step 4: assembling the vector. Both coordinates of $(m_a, m_b)_k = (m_k - u_k, u_k)$ are integer linear combinations of m_k and u_k, so both satisfy the same recurrence, which is equation (266). Its characteristic polynomial is $x^2 - 3m_0 x + 1$; since $m_0 \geqslant 1$ the discriminant $9m_0^2 - 4$ is positive and is not a square. Indeed, if it were d^2, then $(3m_0 - d)(3m_0 + d) = 4$, whose only positive factorisation into factors of the same parity would force $3m_0 = 2$. The dominant root is equation (267), and its other conjugate is

$$\overline{\beta(m_0)} = \frac{3m_0 - \sqrt{9m_0^2 - 4}}{2} = \beta(m_0)^{-1} \in (0,1). \tag{269}$$

Thus $\beta(m_0)$ is an algebraic-integer unit greater than 1 whose only other conjugate has modulus less than 1: it is a quadratic Pisot unit. ☐

Remark 9.7 (Pisot contraction and branch scale)**.** For either coordinate x_k of the vector in equation (266), the exact solution has the form

$$x_k = c_+ \beta(m_0)^k + c_- \beta(m_0)^{-k}, \tag{270}$$

with constants fixed by two consecutive branch values and $c_+ > 0$ on the positive increasing branch. Consequently

$$\log x_k = k \log \beta(m_0) + \log c_+ + O\Big(\beta(m_0)^{-2k}\Big). \tag{271}$$

This is the precise sense in which Pisot contraction gives an exponentially accurate scale along a fixed branch. It does so on every branch, however, and therefore does not distinguish the Fibonacci and Pell arms. Their additional feature is the compatible integral half-step studied below.

The two arms are the cases $m_0 = 1$ and $m_0 = 2$, where $\beta(1) = (3 + \sqrt{5})/2 = \varphi^2$ and $\beta(2) = 3 + 2\sqrt{2} = \lambda^2$.

Remark 9.8 (The order in which $\beta(m_0)$ lives is classical)**.** The discriminant $9m_0^2 - 4$ of equation (267) is not new to Markoff theory. At a fixed largest entry c, the Markoff equation can be written as the pair of identities

$$(a - b)^2 + c^2 = ab(3c - 2), \qquad (a + b)^2 + c^2 = ab(3c + 2), \tag{272}$$

which are printed in Frobenius [9, p. 601 of the reprint] and are quoted as equations (3) and (4) by Zhang [25], and whose cross product is a conic of discriminant $(3c-2)(3c+2) =$

$9c^2 - 4$. That conic is itself classical: in the halved coordinates $A = (a + b)/2$ and $B = (a - b)/2$ it is printed as

$$(2 - 3c)A^2 + (2 + 3c)B^2 = -c^2 \tag{273}$$

in [23, eq. (2.3)]. So the order that carries the branch unit $\beta(m_0)$ is exactly the order attached to m_0 by the classical theory. The two extreme arms provide the two basic instances in which the branch unit has a quadratic-unit square root, although that root need not lie in the order generated by the branch unit. They are also the source of the sharpest known unconditional uniqueness criteria, which run modulo $3c \pm 2$ rather than modulo c: [25, Theorem 2] settles every Markoff number for which one of $3c - 2$, $3c + 2$ is a prime power, four times a prime power, or eight times a prime power, and [23, Theorem 2.3] settles every Markoff number whose $3c - 2$ or $3c + 2$ has a prime power for greatest odd divisor. The two hypotheses coincide on Markoff numbers, because $3c \pm 2$ is odd for odd c, while for even c the quotients $(3c - 2)/4$ and $(3c + 2)/8$ are odd integers [23, Lemma 4.2].

Srinivasan works in exactly the order met here. Her uniqueness criterion [23, Theorem 5.4] asks that every ambiguous form lying in the principal genus of the class group of discriminant $9c^2 - 4$ correspond to a divisor of $3c - 2$, and for a c that is not a prime power she proves that Markoff uniqueness at c is *equivalent* to a statement about $\mathbb{Q}(\sqrt{9c^2 - 4})$ [23, Theorem 5.3]. The order attached to a branch is therefore not an artefact of our construction; it is where the difficulty of the conjecture is already known to sit.

One consequence is worth recording, because it explains why theorem 9.9 is stated for a unit of an arbitrary real quadratic order rather than of this one. The branch unit $\beta(m_0)$ has norm 1 in the order of discriminant $9m_0^2 - 4$, and a square root of it of norm -1 need not lie in that order. At $m_0 = 1$ it does: the discriminant is 5, the order is the maximal order of $\mathbb{Q}(\sqrt{5})$, and φ lies in it. At $m_0 = 2$ it does not: the discriminant is 32, the order is $\mathbb{Z}[2\sqrt{2}]$, and $\lambda = 1 + \sqrt{2}$ lies instead in the maximal order $\mathbb{Z}[\sqrt{2}]$, of discriminant 8. Restricting ε to the order of $\beta(m_0)$ would therefore have discarded the Pell arm. We use none of this, and record it only because it locates proposition 9.6 inside the classical picture.

9.5. **A stringent necessary condition for an integral half-step.** Proposition 9.6 says that a quadratic unit governs every fixed-entry branch. That is not what the proofs in this paper consume. What they consume is the Pell-arm semiconjugacy of proposition 7.1: the map S advances the two-square data by one step, and S^2 realises $\beta(2) = \lambda^2$. An analogous construction at m_0 would therefore need a *square root* of $\beta(m_0)$ acting integrally. The following theorem shows that this is a severe restriction.

For clarity, the square relation is forced by any such fixed action. If $\mathsf{T} \in \mathrm{GL}_2(\mathbb{Z})$ carries every positive two-square vector z_k of a growing fixed-m_0 ray to z_{k+1}, its dominant eigenvalue $\varepsilon > 1$ is a quadratic unit. Since $m_k = \|z_k\|^2$ has growth rate $\beta(m_0)$ by proposition 9.6, while $\|\mathsf{T}^k z_0\|^2$ has growth rate ε^2, necessarily $\beta(m_0) = \varepsilon^2$.

Theorem 9.9 (Trace criterion for a half-step)**.** *Let m_0 be a Markoff number and suppose $\beta(m_0) = \varepsilon^2$ for a unit $\varepsilon > 1$ of a real quadratic order. Then*

$$3m_0 - 2 \textit{ is a perfect square,} \tag{274}$$

and ε has norm -1 and trace $\sqrt{3m_0 - 2}$.

Proof. Let $t = \varepsilon + \overline{\varepsilon}$ be the trace of ε and $N(\varepsilon) = \varepsilon\overline{\varepsilon} = \pm 1$ its norm; both are rational integers. Squaring,

$$\beta(m_0) + \overline{\beta(m_0)} = \varepsilon^2 + \bar{\varepsilon}^2 = t^2 - 2N(\varepsilon). \tag{275}$$

The left-hand side is the trace of $\beta(m_0)$, which is $3m_0$ by equation (267). Hence

$$t^2 = 3m_0 + 2N(\varepsilon). \tag{276}$$

Two cases arise. If $N(\varepsilon) = 1$ then $t^2 = 3m_0 + 2$. But every square is congruent to 0 or 1 modulo 3, while $3m_0 + 2 \equiv 2 \pmod 3$, so this case is impossible. Therefore $N(\varepsilon) = -1$ and $t^2 = 3m_0 - 2$, which is equation (274). □

For the two arms the criterion is satisfied with room to spare: $3 \cdot 1 - 2 = 1$ gives $t = 1$ and $\varepsilon = \varphi$, while $3 \cdot 2 - 2 = 4$ gives $t = 2$ and $\varepsilon = \lambda$. Both have norm -1, as the theorem requires, and both are the units of this paper.

Remark 9.10 (Computational scope)**.** Among all 3502 Markoff numbers below 10^{60}, and again among the 4097 produced by the Christoffel tree to depth 11, the criterion equation (274) is met by exactly three: $m_0 = 1$, $m_0 = 2$ and $m_0 = 34$, with $3 \cdot 34 - 2 = 100$ and $\varepsilon = 5 + \sqrt{26}$. This is a verification on a finite family and not a theorem: we do not know whether $3m - 2$ is a perfect square for infinitely many Markoff numbers m, and nothing below depends on the answer. The certificate is `check_two_arms.py` in the supplement.

The third value can be tested on an explicit ray, and here the calculation is finite because it concerns one concrete linear system.

Proposition 9.11 (No half-step on one displayed fixed-34 ray)**.** *The fixed-34 ray whose successive two-square writings begin* $(5, 8), (34, 89), (513, 814), (3468, 9077)$ *carries no common integral half-step.*

Proof. These successive writings are attached to the Markoff numbers 89, 9077, 925765 and 94418953. A half-step would be a matrix $\mathsf{T} \in \mathrm{GL}_2(\mathbb{Z})$ carrying each writing to the next. The first two conditions already determine T over $\mathbb{Q}$, because $(5, 8)$ and $(34, 89)$ are linearly independent, and solving the resulting linear system gives

$$\mathsf{T} = \frac{1}{173}\begin{pmatrix} -1078 & 1409 \\ 1409 & 1044 \end{pmatrix}, \tag{277}$$

whose entries are not integers. Moreover this T sends $(513, 814)$ to

$$\left(\frac{593912}{173}, \frac{1572633}{173}\right) \neq (3468, 9077), \tag{278}$$

so no matrix at all, integral or not, satisfies the first three conditions simultaneously. □

Within the finite families of remark 9.10, the only non-arm candidate is $m_0 = 34$, and the displayed fixed-34 ray does not support the required action. This is not a classification of all rays fixing 34. Globally, theorem 9.9 is a necessary condition, not a classification: further Markoff values satisfying it are not excluded here. The two arms are nevertheless canonical for the present paper because they are the two constant-directive branches and because their half-step actions are explicit.

9.6. **The two arms are the two constant directive sequences.** There is a second, purely combinatorial way to see the same pair, and it explains where γ comes from.

A Markoff directive word has all its partial quotients in $\{1,2\}$. There are exactly two constant such sequences, and their values are

$$[1;1,1,1,\ldots]=\varphi,\qquad [2;2,2,2,\ldots]=1+\sqrt{2}=\lambda. \tag{279}$$

Recall that a Pisot number is a real algebraic integer greater than 1 whose other conjugates have modulus less than 1, while a Perron number is a real algebraic integer greater than 1 that strictly dominates the moduli of its other conjugates. Both values in equation (279) are Pisot units: their respective conjugates are $-\varphi^{-1}$ and $-\lambda^{-1}$. This does not contrast them with the fixed-entry monodromy eigenvalues of equation (267), which are Pisot on every such branch.

There are in fact three spectral objects: the value of an infinite continued fraction, the dominant eigenvalue of a periodic directive matrix, and the fixed-entry branch unit $\beta(m_0)$. For a constant directive, the continued-fraction value equals the Perron–Frobenius eigenvalue of its one-letter continued-fraction matrix, and its square is $\beta(1)$ or $\beta(2)$. Outside the constant cases the three objects need not coincide. Indeed, put

$$A(a)=\begin{pmatrix} a & 1\\ 1 & 0\end{pmatrix},\qquad \mathsf{P}=A(a_1)\cdots A(a_r)=\begin{pmatrix} A & B\\ C & D\end{pmatrix}. \tag{280}$$

The continued-fraction fixed point $\xi=[\overline{a_1,\ldots,a_r}]$ satisfies

$$C\xi^2+(D-A)\xi-B=0, \tag{281}$$

whose primitive leading coefficient need not be 1. The spectral radius $\rho(\mathsf{P})$ instead satisfies the monic equation

$$X^2-(A+D)X+(-1)^r=0. \tag{282}$$

Its other eigenvalue is $(-1)^r/\rho(\mathsf{P})$, so $\rho(\mathsf{P})$ is a quadratic Pisot unit. For example,

$$[\overline{1,2}]=\frac{1+\sqrt{3}}{2}\quad\text{has primitive polynomial}\quad 2X^2-2X-1, \tag{283}$$

so this continued-fraction value is not an algebraic integer and hence is neither a Pisot nor a Perron number in the arithmetic sense. Its cyclic shift illustrates why the phase matters:

$$[\overline{2,1}]=1+\sqrt{3}\quad\text{has primitive polynomial}\quad X^2-2X-2, \tag{284}$$

and is itself a Pisot number. The two cyclic period matrices have the same Perron–Frobenius eigenvalue $2+\sqrt{3}$, which is a Pisot unit. For a nonperiodic directive the continued-fraction value need not even be algebraic. Thus a Pisot–Perron comparison between branches is meaningful only after the associated number and the continued-fraction phase have been specified. In this paper the relevant branch object is the monodromy eigenvalue $\beta(m_0)$, and the exceptional property of the two units in equation (279) is their compatible half-step action, not Pisotness alone.

These are the Fibonacci and Pell arms. The next proposition converts this into a statement about the cost of a directive word, that is, the number of steps of the subtractive Euclidean algorithm on the two-square pair, which equals the sum of the partial quotients minus one.

Proposition 9.12 (Arm costs and the meaning of γ)**.** *Let $\ell(p,q)$ denote the cost of the pair (p,q). On the Fibonacci arm, where $m=F_{2k+1}$ and $(p,q)=(F_k,F_{k+1})$,*

$$\ell(F_k, F_{k+1}) = k - 1; \tag{285}$$

on the Pell arm, where $m = P_{2k+1}$ and $(p, q) = (P_k, P_{k+1})$,

$$\ell(P_k, P_{k+1}) = 2k - 1. \tag{286}$$

Consequently, comparing the two arms at equal magnitude,

$$\frac{\ell \text{ on the Pell arm}}{\ell \text{ on the Fibonacci arm}} \longrightarrow \frac{2\log\varphi}{\log\lambda} = 2\gamma = 1.09195\ldots \tag{287}$$

Proof. The continued fraction of F_{k+1}/F_k is $[1, 1, \ldots, 1, 2]$ with $k-1$ partial quotients, by induction from $F_{k+1} = F_k + F_{k-1}$; its quotient sum is $(k-2)+2 = k$, so the cost is $k-1$. The continued fraction of P_{k+1}/P_k is $[2, 2, \ldots, 2]$ with k partial quotients, by induction from $P_{k+1} = 2P_k + P_{k-1}$; its quotient sum is $2k$, so the cost is $2k-1$.

For the ratio, Binet's formulas give

$$F_{2k+1} = \frac{\varphi^{2k+1}}{\sqrt{5}} + O(1), \qquad P_{2k+1} = \frac{\lambda^{2k+1}}{2\sqrt{2}} + O(1). \tag{288}$$

Solving each for its index at a common magnitude m gives $k = \log m/(2\log\varphi) + O(1)$ on the Fibonacci arm and $k = \log m/(2\log\lambda) + O(1)$ on the Pell arm. Substituting into the two cost formulas gives $\log m/(2\log\varphi) + O(1)$ and $\log m/\log\lambda + O(1)$ respectively, whose ratio tends to $2\log\varphi/\log\lambda$. ☐

So γ is not a computational artefact of the reduction. It is the asymptotic exchange rate between the directive-word costs of the two arms, and the rational p/Q of equation (125) is a certified rational approximation to that exchange rate.

Remark 9.13 (What is not claimed)**.** It is tempting to strengthen equation (279) into a statement that the two arms are the extremes of the growth rate of admissible directive words. Only one half of that is true. Among all compositions of a given quotient sum N, the continuant is maximal at $(1, 1, \ldots, 1)$ and equals F_{N+1}, so the Fibonacci arm does grow fastest and $\ell(p, q) \geqslant \log(p+q)/\log\varphi + O(1)$ for every writing. The Pell arm, however, is not the slowest: over a common length of twelve letters the Stern–Brocot products for the patterns $(1, 1), (2, 2), (1, 2)$ have traces 322, 198 and 194, so the alternating pattern grows strictly more slowly than the Pell pattern. We therefore make no extremality claim from below.

10. Two-square localisation and the Markoff-facing programme

10.1. **What a two-square writing locates in the Farey tree.** The branch scales answer only a radial question: how large a Markoff value is at a given cost. A two-square writing contains more information. We now make precise when it gives an exact Farey address and when it gives only a candidate.

For coprime integers $0 < p < q$, let $\ell(p, q)$ be the number of single subtractions in the subtractive Euclidean algorithm from (p, q) to $(1, 1)$. Write the canonical continued fraction as

$$\frac{q}{p} = [c_0; c_1, \ldots, c_d], \qquad c_d \geqslant 2. \tag{289}$$

For a normalized Markoff occurrence, write P/Q for its reduced Farey index, with $0 < P < Q$, and $m_{P/Q}$ for its Markoff value.

Proposition 10.1 (Exact Farey recovery and arm benchmarks)**.** *Let $M = p^2 + q^2$ be a primitive positive two-square writing.*

(i) *Its Euclidean cost is*

$$\ell(p,q)=\sum_{j=0}^{d}c_j-1. \tag{290}$$

Recording the side of every subtraction gives, in reverse order, the exact root-to-leaf Stern–Brocot path of q/p, up to exchanging the two coordinates.

(ii) *If the writing is induced by a normalized Markoff occurrence of Farey index P/Q, then*

$$\ell(p,q)=P+Q-2. \tag{291}$$

Its Euclidean half-directive, followed by the inverse Christoffel–Thue–Morse construction, recovers P/Q exactly. An arbitrary primitive writing supplies only a candidate until this validity condition is checked.

(iii) *At cost $L\geqslant 1$, all normalized Markoff occurrences lie on the Farey shell $P+Q=L+2$. The Fibonacci endpoint of this shell is*

$$\frac{1}{L+1},\qquad m_{1/(L+1)}=F_{2L+3}. \tag{292}$$

If $L=2k-1$ is odd, the shell point nearest the diagonal $P=Q$ is the Pell occurrence

$$\frac{k}{k+1},\qquad m_{k/(k+1)}=P_{2k+1}=P_{L+2}. \tag{293}$$

For $L=1$ it coincides with the Fibonacci endpoint. Every occurrence on an odd shell satisfies

$$P_{L+2}\leqslant m_{P/Q}\leqslant F_{2L+3}. \tag{294}$$

If L is even, the formal diagonal point is not reduced and there is no Pell occurrence on that shell. Its lower endpoint is A^*_L/B^*_L, where

$$A^*_L=\max\left\{j<\frac{L+2}{2}:\gcd(j,L+2)=1\right\},\qquad B^*_L=L+2-A^*_L; \tag{295}$$

this is the reduced shell point nearest the diagonal $P=Q$, equivalently the one with largest admissible numerator.

(iv) *Every primitive positive writing of cost L, whether Markoff-valid or not, obeys*

$$M\leqslant F_{2L+3}, \tag{296}$$

with equality only for the unordered Fibonacci pair (F_{L+1},F_{L+2}).

Proof. Grouping consecutive subtractions on the same side gives the ordinary Euclidean quotients $c_0,\dots,c_d$. The final quotient is shortened by one because the subtraction process stops at $(1,1)$ rather than $(1,0)$, which proves equation (290); retaining the sides is precisely the Stern–Brocot itinerary.

For part (ii), use the occurrence construction of Reutenauer and Vuillon [20, Sec. 2 and Theorems 1–2]. If v is the ordinary directive at P/Q, its valid antipalindromic directive is $\theta(\mathrm{Pal}(av))=\widehat{u}u$, where Pal denotes iterated right palindromic closure, $\theta(a)=ab$, $\theta(b)=ba$, and $\widehat{u}$ denotes reversal followed by the exchange $a\leftrightarrow b$. To keep the two-square step self-contained, let

$$L_a=\begin{pmatrix}1&1\\0&1\end{pmatrix},\qquad L_b=\begin{pmatrix}1&0\\1&1\end{pmatrix},\qquad \mathbf{e}=\begin{pmatrix}1\\1\end{pmatrix} \tag{297}$$

be the incidence matrices and put $L_u\mathbf{e} = (r,s)^\mathsf{T}$. Since $L_{\widehat{u}} = L_u^\mathsf{T}$, the Christoffel incidence formula and the cited occurrence construction give

$$M = \mathbf{e}^\mathsf{T} L_{\widehat{u}u}\mathbf{e} = \mathbf{e}^\mathsf{T} L_u^\mathsf{T} L_u\mathbf{e} = r^2 + s^2. \tag{298}$$

By the hypothesis of part (ii), the unordered pair $\{r,s\}$ is $\{p,q\}$. Thus the ordered occurrence count vector is either (p,q) or (q,p); exchanging the coordinates complements u. Consequently subtractive Euclid on the sorted pair recovers u up to this orientation and gives $\ell(p,q) = |u|$. The normalized convention selects the orientation for which the inverse image below begins with the prescribed letter a. Since θ doubles length and the displayed antipalindrome has two equal halves,

$$|u| = |\mathrm{Pal}(av)|. \tag{299}$$

The incidence formula associated with the Farey index P/Q gives $|a\,\mathrm{Pal}(av)b| = P+Q$. Removing the boundary letters proves equation (291). The Thue–Morse block code and finite palindromization are injective, and proper lower Christoffel words are in bijection with reduced Farey indices. Thus reversing the normalized construction recovers P/Q.

Part (iii) now begins with $P + Q = L + 2$. The fixed-sum ordering theorem [12, Proposition 22] says that the Markoff values strictly decrease as P moves from 1 towards the diagonal $P = Q$. The value at $1/(L+1)$ is the Fibonacci-arm value F_{2L+3}. When $L = 2k-1$, the shell sum is $2k+1$, so its central reduced point is $k/(k+1)$, whose value is the Pell-arm value P_{2k+1}. This proves equation (294). When L is even, $L+2$ is even and the diagonal point has equal numerator and denominator, so it is not a normalized reduced index. The largest admissible numerator below the diagonal is exactly A_L^* of equation (295), and fixed-sum monotonicity makes its fraction the lower endpoint.

Finally, reverse the subtraction process from $(1,1)$. After j additions, sorting the coordinates as $x_j \leqslant y_j$ gives inductively

$$x_j \leqslant F_{j+1}, \qquad y_j \leqslant F_{j+2}. \tag{300}$$

At $j = L$ this yields

$$M = p^2 + q^2 \leqslant F_{L+1}^2 + F_{L+2}^2 = F_{2L+3}. \tag{301}$$

Equality in both coordinate bounds forces the Fibonacci choice at every nonsymmetric addition, proving the equality statement in part (iv). □

This proposition gives a sharp answer to two superficially similar localisation questions. If the valid writing is known, no approximation is needed: Euclid gives ℓ, the full mixed itinerary, and the Farey point exactly. If only the integer M is known, its primitive two-square writings give candidate itineraries. Proving that the occurrence-induced one is never beaten in cost is the unresolved weak-minimality step and cannot be supplied by a size comparison without resolving part of the Markoff uniqueness problem.

Remark 10.2 (Enumerating the candidate tree positions)**.** For a specified factored integer $M > 2$ admitting a primitive positive two-square writing, primitivity forces

$$M = 2^\epsilon \prod_{j=1}^{r} p_j^{e_j}, \qquad \epsilon \in \{0,1\}, \qquad p_j \equiv 1 \pmod 4, \tag{302}$$

where the p_j are distinct. Choose a Gaussian prime π_j above each p_j. Unique factorisation in $\mathbb{Z}[i]$ shows that a primitive Gaussian integer of norm M contains, for every j, either the complete block $\pi_j^{e_j}$ or the complete conjugate block $\overline{\pi_j}^{e_j}$. Splitting one block between the two conjugates would make p_j divide both coordinates. Conversely, every choice of one complete block gives a primitive writing. Quotienting by Gaussian units and global conjugation therefore gives exactly

$$2^{r-1} \tag{303}$$

normalized unordered primitive positive writings of M.

Part (i) of proposition 10.1 sends each writing to its complete subtractive-Euclidean path. The inverse Christoffel–Thue–Morse construction then tests whether that path has the required occurrence form, and a Cohn evaluation checks the proposed value. Thus every possible position arising from a writing of this fixed M is explicitly enumerable.

This procedure is a localization algorithm, not the missing obstruction. It depends on the factorisation of the particular M, gives no uniform bound on the number or geometry of the candidate paths, and does not prove that two distinct valid candidates cannot survive for some larger value. The global problem is to compare or exclude those candidates without enumerating one integer at a time; weak minimality is one sufficient mechanism.

The relevance of that selection problem to the ultimate programme can be stated as a short conditional reduction.

Proposition 10.3 (Reduction to weak two-square minimality)**.** *Assume that, for every normalized non-endpoint Markoff occurrence, its occurrence-induced pair (p, q) minimizes $\ell(p, q)$ among all primitive positive writings of the same Markoff value as a sum of two squares. Ties are allowed. Then the Markoff–Frobenius uniqueness conjecture holds.*

Proof. Suppose two normalized occurrences have the same Markoff value M, and let (p, q) and (p', q') be their induced writings. Applying the assumed minimality first to one occurrence and then to the other gives

$$\ell(p, q) \leqslant \ell(p', q') \qquad \text{and} \qquad \ell(p', q') \leqslant \ell(p, q). \tag{304}$$

The costs are equal. By equation (291), the two Farey indices P/Q and P'/Q' therefore have the same numerator–denominator sum. If they were distinct, the strict fixed-sum ordering [12, Proposition 22] would give distinct Markoff values, a contradiction. Hence the normalized Farey occurrences coincide. The boundary values 1 and 2 are immediate. □

The proposition deliberately assumes only non-strict minimality. It is not a theorem of the present paper that the occurrence-induced writing has this property. Rather, proposition 10.3 identifies a precise sufficient upstream statement that would turn the two-square/Farey dictionary into a proof of Markoff uniqueness. The arm estimates below explain why scale alone does not yet establish that statement.

The arm points nevertheless calibrate the scale. Solving equations (292) and (293) asymptotically for L gives

$$L_{\mathrm{F}}(M) = \frac{\log M}{2 \log \varphi} + O(1), \qquad L_{\mathrm{P}}(M) = \frac{\log M}{\log \lambda} + O(1), \tag{305}$$

and their ratio tends to $2 \log \varphi / \log \lambda = 2\gamma$. On odd valid shells these are genuine endpoint scales by equation (294); for arbitrary run-$\{1, 2\}$ directives the Pell scale is not a lower envelope, as remark 9.13 shows. Whenever the two arm estimates furnish a genuine valid-occurrence bracket, their multiplicative separation leaves an additive uncertainty proportional to $\log M$, not an $O(1)$ determination of L. For arbitrary run-$\{1, 2\}$ directives, only the Fibonacci side is a proved universal envelope. A fine estimate therefore requires additional branch information; a prefix of the actual Farey path is one natural choice. Such a prefix gives a Stern–Brocot cylinder. If the continuation thereafter follows one fixed-entry branch, the Pisot estimate of remark 9.7 controls its periodic scale exponentially well; the prefix alone neither chooses that continuation nor supplies one fixed branch unit.

10.2. **Dictionary.** The correspondence between the word-combinatorial objects and the objects of this paper is exact, and is summarised in table 1. Its first row is an identity rather than an analogy: by corollary 9.3 the letter counts are read off the writing by a congruence, and exchanging the order of the writing exchanges the two letter counts.

TABLE 1. The word-combinatorial reading of the objects of this paper.

Christoffel–Markoff side	This paper
valid occurrence writing $m = p^2 + q^2$	seed $z = (u, v)$ with $\mathcal{H}(z) = u^2 + v^2$
one step along the Pell arm	$\mathsf{S}(u, v) = (v, u + 2v)$
that step, in the unit	multiplication by λ
Fibonacci-arm writing (F_{q+1}, F_{q+2})	plus core $z_{q,+}$
Fibonacci/Pell arm letter counts	coordinates of φ^{2k} or λ^{2k}
Fibonacci/Pell arm cut values	the coefficients F_r, or C_r and P_r
Fibonacci-arm cost	q for the plus core $z_{q,+}$
Pell-arm cost	$2k - 1$ for (P_k, P_{k+1})

Equation (64) is the instance of this dictionary used by the main proof.

10.3. **Consequence for the Markoff-facing programme.** Theorem 9.9 and proposition 9.11 show why a branchwise extension is not automatically obtained by changing a parameter. Given the general orbit/gap equivalence of theorem 7.2, it is natural to seek an analogue of theorem 1.1 on another branch by transporting S. Any candidate must first have $3m_0 - 2$ square and must then admit a compatible integral action on its two-square sequence. The value $m_0 = 34$ passes the first test, while the displayed fixed-34 ray fails the second. Other fixed-34 rays and further square-condition values are not globally excluded. The general monodromy recurrence alone supplies neither the half-step nor the norm–content geometry, so an extension needs a new compatibility mechanism, not merely a new growth unit.

11. Scope, connections, and the next obstruction

The first arithmetic conclusion of this paper is complete on its stated domain: for every $q \geqslant 1$, both signs, and every positive exponent, the strict Fibonacci–Pell norm–content system has exactly the two rows in equation (7). The proof is unbounded in q and closes only after the analytic bounds have reduced both parity branches to exhaustive finite domains. The second conclusion is likewise complete: for both signed Fibonacci cores, every positive Pell anchor, and every nonnegative time, theorem 7.4 gives all orbit hits.

The surrounding results explain why these classifications are rigid. At the Archimedean place, candidate windows are abundant and their exact densities are Haar measures of interval intersections on a phase circle. At split finite places, compatible Fibonacci and companion ranks can synchronise to arbitrary prescribed depth. Neither phenomenon sees the oriented primitive unit direction. The global norm–content equation sees both coefficient coordinates simultaneously, and it is this global orientation that enables the primitive rectangle and the logarithmic-form argument. The memorable separation is therefore not local versus computational, but local and measure-theoretic abundance versus global Diophantine rigidity.

The algebraic structures exposed here have precise roles. The norm torus, its negative-norm torsor, the integral unit group, and the Lorentzian symmetric-square representation all act directly in the proofs. The biquadratic field $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ supplies the two conjugations used in the norm and height arguments. These are the cross-field structures needed by the present proof. Further frameworks would become relevant only after an explicit map or invariant connects them to the classified system.

There is, however, a genuine algebraic-geometric object behind the parent problem. Over $\mathbb{Z}$, the Markoff equation defines the affine cubic scheme

$$\mathcal{M} = \operatorname{Spec} \mathbb{Z}[a, b, c]/(a^2 + b^2 + c^2 - 3abc), \tag{306}$$

and the Vieta involutions, such as

$$(a, b, c) \longmapsto (a, b, 3ab - c), \tag{307}$$

are integral automorphisms of this cubic. After the scaling $(x, y, z) = (3a, 3b, 3c)$, its equation becomes $x^2 + y^2 + z^2 - xyz = 0$, the Fricke trace surface with commutator trace -2. This description is also a concrete algebraic-topological bridge. The fundamental group of a one-holed torus is the free group F_2; the three coordinates are traces of two generators and their product in an SL_2-representation. Goldman shows that the reflection

$$(x, y, z) \longmapsto (x, y, xy - z)$$

comes from the extended mapping-class group of the surface [10, Sections 2.1–2.2 and Appendix A-3]. Scaling back gives exactly equation (307). On the discrete side, regarding positive Markoff triples as vertices and Vieta moves as edges gives the Markoff mutation graph and its familiar descent tree; Christoffel words and integral necklaces give combinatorial encodings of its arithmetic vertices [19, 8]. Thus topology explains the symmetry, graph theory records its orbits, and combinatorics labels the vertices. These are exact structural connections, but not hidden inputs: this paper constructs no morphism from a hypothetical collision locus on $\mathcal{M}$ to the Fibonacci nearest-gap system.

The most immediate arithmetic extension starts instead from theorem 7.2. For every noncanonical positive seed, an orbit hit is equivalent to a strict norm–content gap for the seed-dependent quadratic integer A_z. The Fibonacci cores form one coefficient locus on which the present paper achieves a complete classification. Noncanonical cores with $\mathcal{Q}(z)^2 > 1$ lead to more general continuant-dependent coefficients. The next problem is to reconstruct enough primitive geometry from those coefficients to retain effective heights without assuming the very collision one hopes to exclude. That problem is suitable for a separate paper because it requires new arithmetic, not additional exposition around the two-row theorem.

The moving-coefficient logarithmic form should also be understood at the level of mechanism. Grouping all terms of the same dominant growth before applying a logarithmic-form estimate is established practice in exponential Diophantine equations [6, 4]. What is specific here is the coefficient equation (97), its exact full norm equation (105), and the certified reduction that turns those identities into a complete classification. We make no priority claim for the general strategy.

The Markoff-facing bottleneck is therefore sharp. Downstream of the Fibonacci-core hypothesis, the gaps and all positive-anchor orbits are classified. Upstream, one must prove that a repeated largest Markoff number yields one of these cores, find a competing core, or classify the seed-dependent gaps with $\mathcal{Q}(z)^2 > 1$.

One tempting fourth route meets a sharp obstruction. The map S is an integral square root of the monodromy Pisot unit $\beta(2) = \lambda^2$ of the Pell arm, and theorem 9.9 shows that an analogous half-step on the branch through m_0 can exist only when $3m_0-2$ is a perfect square. One displayed ray fixing $m_0 = 34$ shows that this arithmetic condition alone does not guarantee compatibility: its two-square data admit no common integral half-step. Other fixed-34 rays are not classified here. Transport to another branch would require both the square condition and a compatible action on that ray's two-square sequence; neither compatibility nor its failure is supplied by the general branch recurrence alone.

The frontier is precise: local abundance versus global rigidity. Progress on Markoff uniqueness requires crossing the missing upstream arrow in figure 1.

Appendix A. Certificate architecture and finite coverage

This appendix specifies exactly where computation enters the proof and how the finite conclusions cover the original unbounded statement. The corresponding source files and one-command driver are listed in the supplement, and table 2 records which certificate carries which claim.

Table 2. Computer-assisted inputs and controls. Finite density and orbit regressions test identities and boundaries but do not prove the corresponding unbounded theorems.

Claim	Complete input	Independent control
Leading exponents	Published complete Fibonacci–Pell and Pell near-square classifications	Two optional Magma V2.29-9 quartic runs and exact arithmetic filters
Even classification	Exact logarithmic-form identities, outward rational intervals, all 190 moving shifts, and enumeration of $2 \leqslant q < 90$	Two separately structured fixed-point implementations and a wider exact regression
Odd exclusion	Exact changed-sign trace identities, unchanged fixed and moving shifts, and enumeration of $q = 3, 6, \ldots, 93$	A source checker and a non-importing integer-arithmetic reconstruction
Window densities	Irrational-rotation proof and exact interval lengths	Exact finite counts through $q = 5000$, used only as regression evidence
All-anchor orbits	Exact semiconjugacy, anchored equivalence, all-exponent theorem, and published direct-hit classifications	Separate general-seed, odd-anchor, and even-anchor identity regressions
Uniform local clocks	Written p-adic rank, Chinese remainder, and equidistribution proof	Exact rank certificates, obstruction at 241, and displayed witnesses

A.1. **Logical dependency.** No finite computation proves an unbounded statement by extrapolation. In the even branch, the analytic argument applies to every hypothetical target with $q \geqslant 7$: propositions 3.2 and 4.3 and the moving-coefficient form first give $q < 10^{32}$; the certified reductions then give $h < 192$ and $q < 90$. Only after those deductions does the terminal enumeration inspect the complete finite domain

$$2 \leqslant q < 90, \qquad 3 \nmid q, \qquad \sigma \in \{1, -1\}. \tag{308}$$

The omitted lower boundary is analytic rather than computational: equation (142) proves directly that neither sign has a strict candidate at $q = 1$. Thus $q = 90$ is excluded by the strict analytic bound, while every lower index in this parity branch and both signs are covered. In the odd branch, the fixed-form-first argument gives $q < 10^{32}$, the fixed reduction gives $\min\{h, q\gamma\} < 48$, and the moving reduction excludes $q \geqslant 96$. It leaves exactly

$$q = 3, 6, \ldots, 93, \qquad \sigma \in \{1, -1\}. \tag{309}$$

Exact recurrence arithmetic checks both complete domains. The even output consists only of equations (143) and (144), while the odd output is empty. This proves both existence and completeness of theorem 1.1.

A.2. **Rational interval model.** All reduction comparisons are made with rational endpoints. Positive square roots are bracketed by integer squaring. After range reduction, logarithms are evaluated through equations (140) and (141). A certificate is accepted only when the lower endpoint of each κ is positive and the lower endpoint of the left side in equations (129) and (137) exceeds the upper endpoint of its right side. Strict inequalities therefore remain strict under rounding.

The common integers p, Q are printed in equation (125). The certificate independently verifies $\gcd(p, Q) = 1$ and the approximation error; trust in a floating-point continued-fraction library is not part of the proof. The moving stage evaluates every one of the 190 pairs, rather than sampling or replacing them by a numerical minimum.

A.3. **Exact arithmetic model.** An element of $\mathbb{Z}[\sqrt{2}]$ is stored as a pair of arbitrary-precision integers (a, b) representing $a + b\sqrt{2}$. Addition, multiplication, conjugation, norm, and coefficient content are consequently exact. The sign of $a + b\sqrt{2}$ is decided from integer signs and the comparison of a^2 with $2b^2$. Fibonacci and Pell coordinates are generated by their integer recurrences. The terminal search therefore contains no floating-point predicate.

The two all-exponent programs share the published classification inputs and the fixed rational-interval constants already displayed in the manuscript. The second independently reconstructs the parity selector, the changed odd trace, the bootstrap arithmetic, and the terminal integer enumeration; it does not import the first. Wider runs through $q = 2000$ are regressions only.

A.4. **Published near-square classification.** The proof of equation (62) imports the published complete classification in [3, Table 1]. That paper reduces the relevant Lucas near-square equations through complete computational reductions via elliptic curves or Thue equations. Because its table does not publish row-specific transcripts, the supplement retains a separate complete calculation of the two induced quartics. The local Magma transcripts report the complete integral-point routine and the required Mordell–Weil information; standard-library programs then verify every returned point and the exact filtering to equation (62). The manuscript relies on the published theorem for completeness and treats the local calculation as independent corroboration.

A.5. **Reproduction.** From the manuscript directory, run

```
python3 supplement/run_all.py
```

This executes every standard-library certificate used by the paper. Magma is optional: its inputs and recorded transcripts can be inspected without the proprietary executable, while a fresh Magma V2.29-9 installation can rerun them. The supplement README records the software assumptions, source-to-claim map, expected summaries, and the distinction between proof certificates and non-probative regression checks.

Code availability

The LaTeX sources, supplementary proof certificates, recorded reference outputs, and reproduction instructions accompanying this article are publicly available in the companion GitHub repository:

https://github.com/dutykh/fibonacci-pell-nearest-gaps/

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